

A Hybrid Monte Carlo-Deterministic Second Moment Method with Efficient Variance Reduction

Michael Pozulp^{a,*}, Terry Haut^b, Patrick Brantley^c, Samuel Olivier^d

^a*Applications, Simulations, and Quality Division, Lawrence Livermore National Laboratory, Livermore, California 94550*

^b*Center for Applied Scientific Computing, Lawrence Livermore National Laboratory, Livermore, California 94550*

^c*Design Physics Division, Lawrence Livermore National Laboratory, Livermore, California 94550*

^d*Computer, Computational, and Statistical Sciences Division, Los Alamos National Laboratory, Los Alamos, New Mexico 87545*

Abstract

We present a hybrid method that combines Monte Carlo with deterministic finite element methods to solve a linear Boltzmann transport equation. Our hybrid method runs orders of magnitude faster than Monte Carlo, without sacrificing accuracy, for a proxy problem from radiative transfer that contains both optically-thick and optically-thin material. We believe that this is the first demonstration of a hybrid Second Moment Method in more than one spatial dimension, the first to consider more than one material, and the first to use variance reduction. Our variance reduction approach arises from an asymptotic analysis in which we show that the magnitude of the scattering source grows without bound. We transform the problem to compute the deviation of the radiation intensity from isotropy. The magnitude of the source in the transformed problem is bounded, and the quality of the hybrid method solution is dramatically improved by a substantial reduction in the variance.

Keywords: Second Moment Method, Monte Carlo, Variance Reduction

1. Introduction

The Second Moment Method (SMM) is an iterative method which solves a linear Boltzmann transport equation and a system of linear moment equations every iteration cycle [1]. The transport equation and the moment equations are coupled through source terms. The linearity and source coupling of SMM contrasts with the non-linearity and operator coupling of Quasidiffusion, or Variable Eddington Factor (VEF) [2]. SMM and VEF converge quickly in the optically-thick, highly-scattering transport regime known as the thick diffusion limit (TDL) [3]. Source iteration (SI), which lags the scattering source in order to avoid forming the transport operator, converges arbitrarily slowly in the TDL. Methods such as diffusion synthetic acceleration (DSA) accelerate SI convergence [4]. SMM and VEF are similar to DSA, in the sense that all three methods converge quickly, but there are some important differences. For example, the low-order problem must be “consistent” with the high-order problem for DSA to always converge quickly [5].

Our hybrid method lags the scattering source, not to avoid forming the transport operator, but to avoid simulating scattering events in the Monte Carlo particle histories. Simulating these events can dominate the calculation runtime in the TDL. However, just like SI with deterministic methods, Monte Carlo SI converges arbitrarily slowly in the TDL. We fix this problem by creating a hybrid SMM that converges quickly. We use SMM instead of DSA because our use of Monte Carlo to solve the high-order problem makes it unclear how we would satisfy the consistency requirement. We chose SMM over VEF because of statistical variation which could be more complicated to manage when it appears in the operator, as in hybrid VEF, rather than in the source, as in hybrid SMM. We use Monte Carlo because it allows the angular dimensions of

*Corresponding author

the high-order problem to remain continuous, thereby avoiding discretization errors such as the unphysical starburst patterns known as “ray effects” that are caused by angular under-resolution.

Using Monte Carlo (MC) methods yields unbiased estimators, but finite sampling introduces statistical variation—hereafter referred to as noise—that can reduce their precision. Hybrid methods may amplify this noise through differentiation. We avoid noise amplification by solving the first-order form of the low-order problem, which involves only a single spatial derivative acting on the noisy source term, and we use integration by parts with a finite element method to transfer the derivative from the noisy source term to a test function in the weak formulation. Although the interior face terms in the weak form could be calculated using surface estimators, we instead use volumetric estimators and apply finite element interpolation to evaluate these terms in the weak form. We observed numerical evidence suggesting that solving the weak, first-order form without surface estimators is sufficient for avoiding noise amplification.

However, we found that simply avoiding noise amplification is not sufficient to provide an efficient hybrid alternative to Monte Carlo for calculations in the previously mentioned TDL regime. To address this issue, we perform an asymptotic analysis to derive an expression relating the variance of the MC estimators in our hybrid method to ε , the parameter characterizing the TDL. Our analysis, which uses a variance result from Pozulp and Haut [6], demonstrates that the variance of path length estimators for single-collision Monte Carlo source iteration methods such as ours is unbounded in the TDL. Therefore, we develop and implement a source transformation variance reduction technique that transforms the high-order problem into one that requires computing the deviation of the radiation intensity from isotropy. In the transformed problem, the variance of the MC path length estimators is bounded, rather than unbounded.

The transformed problem resembles the difference formulation of Szoke and Brooks [7], a transport reformulation in which one solves for the difference between the intensity and a prescribed reference field (often the Planck distribution in thermal radiative transfer), rather than for the intensity itself. As a result, the material emission source is replaced by the space-time derivative of the Planck function. The spatial gradient term in the inhomogeneous source of the difference formulation can be addressed by sourcing MC particles on the interior faces of the mesh [8]. A notable feature of our hybrid method is that we do not use interior face sources, despite a similar gradient appearing in our inhomogeneous source term. Although this approach is possible within our hybrid method—and would require sourcing only on the interior mesh faces that coincide with material interfaces—we instead apply finite element differentiation to compute a volumetric source which we sample instead of face sources. Our technique is an extension of the difference formulation to iterative Monte Carlo, and the resulting equations in the high-order problem resemble those in the generalized difference formulation [9].

An example of an issue that our hybrid method could fix, were it to be extended to nonlinear transport, is the very long runtimes of Monte Carlo calculations which solve the thermal radiative transfer equations using Fleck & Cummings’ “implicit Monte Carlo” (IMC) linearization [10]. The IMC linearization produces a linear transport equation with an effective scattering source which causes arbitrarily long IMC photon histories in the TDL, leading to arbitrarily long IMC calculation runtimes. Fleck & Canfield’s Random Walk (RW) algorithm can reduce the runtime by replacing many steps with a single step in the histories of IMC photons which satisfy certain properties [11]. The equation that we solve is linear, unlike thermal radiative transfer, but it still reproduces the long IMC runtime issue.

Runtime improvements exceeding that of RW can be achieved using more recently developed IMC acceleration algorithms, such as implicit Monte Carlo Diffusion (IMD) [12, 13] or discrete diffusion Monte Carlo (DDMC) [14, 15]. IMD and DDMC are very effective, but not perfect. Both algorithms use diffusion approximations, require phase-space partitioning, and have not been extended to meshes with reentrant elements. In contrast, our method does not use diffusion, does not partition phase-space, and does not have issues with reentrant elements. Re-entrant (concave) elements can arise in Lagrangian radiation-hydrodynamics due to mesh distortion, such as when initially convex quadrilaterals become concave under strong shear [16].

Although we suggest thermal radiative transfer as an application of our method, we make no assumptions about the type of particle in the simulation—our method is applicable as long as the linear transport equation under consideration is a valid physical model for the particle type. Neutrons, like photons, can be simulated in accordance with a linear transport equation. One example of a problem that our hybrid method could address in neutronics is the high computational cost of Monte Carlo transport in the moderator surrounding

the core in a nuclear power reactor, where neutrons undergo many scattering collisions in optically thick, weakly absorbing media [17], leading to long particle histories and slow runtimes [18].

SMM and VEF algorithms typically use deterministic methods to solve both the high-order transport problem and the low-order moment system. We use the term “hybrid” because our method employs Monte Carlo to solve the high-order problem. We believe that our method is the first demonstration of a hybrid SMM in more than one spatial dimension, the first to consider more than one material, and the first to use variance reduction. However, a few similar hybrid methods exist. These can be grouped into two categories: those that interrupt a Monte Carlo calculation one or more times to solve a moment system without actually iterating, and those that iterate to converge quantities such as a source term or the Eddington tensor.

The “interrupt” category of similar hybrid methods contains what may be the earliest example of a hybrid SMM or VEF method: Cooper & Larsen’s method to compute weight windows to reduce the variance of a Monte Carlo calculation [19]. By interrupting their MC calculation to solve a VEF system, then using the VEF system solution to set weight windows before resuming their MC calculation, Cooper & Larsen reduced the variance of their MC solution, while also improving an important Monte Carlo figure of merit. More recently, Novellino & Anistratov computed MC solutions for a transport problem, followed by SMM and VEF system solutions, and found that the moment system solutions were sometimes more accurate than the MC solution [20].

The category of similar hybrid methods which iterate is larger than the “interrupt” category. Willert et al. replaced the deterministic transport solve in a method that uses nonlinear diffusion acceleration (NDA) with Jacobian-free Newton-Krylov (JFNK) to accelerate source iteration in a method called JFNK-NDA(MC) [21]. Park et al. created a hybrid method by using Monte Carlo with the moment method that they described in [22]. In their hybrid method, Park et al. use MC to compute consistency terms which appear in their moment system [23]. They use a phase-space partitioning technique, described in more detail by Densmore et al. [24], which is similar to that of IMD and DDMC. Lam et al. solved the thermal radiative transfer equations by using a transport-corrected diffusion solve to compute the effective scattering source. They lag this source and iterate between single-collision Monte Carlo transport and deterministic inversion of a diffusion operator until the effective scattering source in the transport equation converges [25]. Pasmann et al. used a low-discrepancy sequence instead of a pseudo-random number sequence to improve the asymptotic order of the estimator uncertainty such that a factor of two reduction in the uncertainty requires only twice as many particles, instead of four times as many. This enhancement violates assumptions which allow for the contribution of the scattering and fission sources to be included through scattering events and fission events. Pasmann et al. accounts for the contribution in the particle weights, and re-computes the contribution every cycle of an iteration [26]. Lam et al. were also able to use a low-discrepancy sequence because their method accounts for the contribution of the effective scattering source in the particle weights, rather than including effective scattering events in the particle histories [25].

We refer to our method as hybrid second moment (HSM), and in section 2 we describe HSM without any derivations. The purpose of section 2 is to summarize the method before showing details, such as derivations, which we include in subsequent sections. Derivations for the deterministic and Monte Carlo components of the hybrid method appear in sections 3 and 4, respectively. Section 5 provides an analysis of the MC estimator variance in the TDL. Finally, we show numerical results in section 6 which demonstrate that HSM is orders of magnitude faster than Monte Carlo, without sacrificing accuracy, for a proxy problem from radiative transfer that contains both optically-thick and optically-thin material. We conclude with a summary and suggestions for future work in section 7.

2. Hybrid Second Moment Algorithm

A steady-state, frequency-independent, linear Boltzmann transport equation may be written,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \int_{\mathbb{S}^2} \psi \, d\Omega' + q, \quad (1a)$$

where $\boldsymbol{\Omega} \in \mathbb{S}^2$ is the unit direction vector, $\mathbb{S}^2$ is the unit sphere, $\psi = \psi(\mathbf{x}, \boldsymbol{\Omega})$ is the radiation intensity, $\mathbf{x} \in \mathbb{R}^{\text{dim}}$ ($\text{dim} = 1, 2, 3$) is the spatial position coordinate in a Cartesian spatial geometry, σ_t is the total

material opacity, σ_s is the scattering opacity, q is an arbitrary source function, and $d\Omega'$ is the differential solid angle element on the unit sphere $\mathbb{S}^2$ associated with the integration variable Ω' so that $\int_{\mathbb{S}^2}(\cdot) d\Omega'$ is an integral over all directions. Equation (1a) is subject to an inflow boundary condition, which defines the radiation intensity on the domain boundary in the hemisphere of directions that point into the domain at a given location on the domain boundary surface,

$$\psi(\mathbf{x}, \boldsymbol{\Omega}) = \psi_{\text{inc}}(\mathbf{x}, \boldsymbol{\Omega}), \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \boldsymbol{\Omega} \cdot \mathbf{n} < 0, \quad (1b)$$

where ψ_{inc} is an arbitrary inflow function, $\partial\mathcal{D}$ is the domain boundary, and $\mathbf{n}$ is the outward facing unit normal vector on the domain boundary surface. The second moment system associated with the linear transport problem can be derived by taking angular moments of eq. (1a). A boundary condition for the system can be derived by manipulating partial currents. The second moment system is (see section 3 for the derivation),

$$\nabla \cdot \mathbf{J} + \sigma_a \varphi = Q_0, \quad \mathbf{x} \in \mathcal{D}, \quad (2a)$$

$$\frac{1}{3} \nabla \varphi + \sigma_t \mathbf{J} = \mathbf{Q}_1 - \nabla \cdot \mathbf{T}, \quad \mathbf{x} \in \mathcal{D}, \quad (2b)$$

$$\mathbf{J} \cdot \mathbf{n} = \frac{1}{2} \varphi + 2J_{\text{in}} + \beta, \quad \mathbf{x} \in \partial\mathcal{D}, \quad (2c)$$

where $\mathbf{J} \in \mathbb{R}^{\text{dim}}$ is the vector unknown, $\sigma_a = \sigma_t - \sigma_s$ is the absorption opacity, φ is the scalar unknown, Q_0 is the zeroth angular moment of q , $\mathbf{Q}_1$ is the first angular moment of q , J_{in} is a functional of ψ_{inc} , and $\mathbf{T}$ and β functionals of ψ . The functionals are,

$$J_{\text{in}}(\psi_{\text{inc}}) = \int_{\boldsymbol{\Omega} \cdot \mathbf{n} < 0} \boldsymbol{\Omega} \cdot \mathbf{n} \psi_{\text{inc}} d\Omega, \quad (3)$$

$$\mathbf{T}(\psi) = \int_{\mathbb{S}^2} \boldsymbol{\Omega} \otimes \boldsymbol{\Omega} \psi d\Omega - \frac{1}{3} \mathbf{I} \int_{\mathbb{S}^2} \psi d\Omega, \quad (4)$$

$$\beta(\psi) = \int_{\mathbb{S}^2} |\boldsymbol{\Omega} \cdot \mathbf{n}| \psi d\Omega - \frac{1}{2} \int_{\mathbb{S}^2} \psi d\Omega, \quad (5)$$

where $\mathbf{I}$ is the identity matrix. When $\mathbf{T}$ and β are zero, the system of eqs. (2a) and (2b) simplifies to the radiation diffusion approximation, and eq. (2c) simplifies to the Marshak boundary condition [17]. For this reason, we refer to $\mathbf{T} \in \mathbb{R}^{\text{dim} \times \text{dim}}$ as the transport correction tensor and β as the boundary correction factor. While $\mathbf{T}$ has nine entries when $\text{dim} = 3$, only six are unique, because $\mathbf{T}$ is symmetric. Before discretization, eqs. (2a) to (2c) are an equivalent reformulation of eqs. (1a) and (1b). Thus, before discretization, the scalar and vector unknowns in the SMM system are equal to the zeroth and first angular moments of the intensity. By construction, the distribution associated with the moment solution is the distribution that exactly satisfies the transport equation. We made no assumptions, nor approximations, in deriving the moment system.

The Second Moment Method uses φ , which comes from solving the moment system, to compute the scattering source in eq. (1a). We write this by simply replacing the angle integrated intensity in eq. (1a) with φ ,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \sigma_t \psi = \frac{\sigma_s}{4\pi} \varphi + q. \quad (6)$$

Solving eq. (6) along with the boundary condition eq. (1b) for ψ allows us to compute the SMM data $\mathbf{T}$ and β , which are functionals of ψ . Figure 1 shows the SMM iteration. In our hybrid method, we may use Monte Carlo without scattering events to solve the linear transport equation (left side of fig. 1) and compute the SMM data $\mathbf{T}$ and β as Monte Carlo estimators. We may then solve the moment system (right side of fig. 1) using a deterministic method, and use the solution φ to compute the scattering source, which we converge in an iteration.

However, such a hybrid method would have a Monte Carlo solution estimator with an arbitrarily large variance in the TDL. Specifically, $\text{Var}[\cdot] \sim 1/\varepsilon$, where $\varepsilon \in (0, 1]$ is a dimensionless parameter by which the

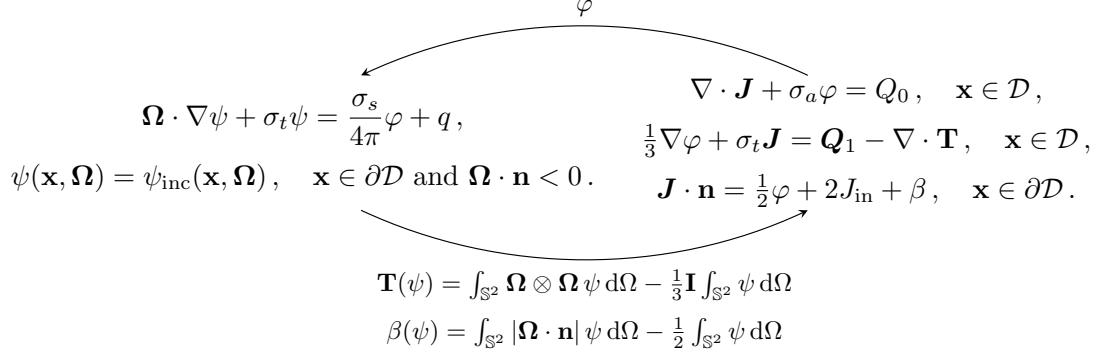


Figure 1: SMM algorithm.

problem data are scaled such that taking the limit $\varepsilon \rightarrow 0$ of the scaled problem induces the TDL [3]. To fix the noise issue, we must fix the undesirable dependence of the variance on ε . We do this by substituting $\psi = \bar{\varphi}/(4\pi) + \tilde{\psi}$ into eq. (6), where $\tilde{\psi}$ is a new unknown quantity representing the amount by which the intensity ψ deviates from $\bar{\varphi}/(4\pi)$, and $\bar{\varphi}$ is an arbitrary function which we leave undefined for now. The result of the substitution is,

$$\Omega \cdot \nabla \tilde{\psi} + \sigma_t \tilde{\psi} = \frac{\sigma_t}{4\pi} (\varphi - \bar{\varphi}) - \frac{1}{4\pi} (\sigma_a \varphi + \Omega \cdot \nabla \bar{\varphi}) + q. \quad (7a)$$

Making the same substitution into eq. (1b) gives the new inflow boundary condition,

$$\tilde{\psi}(\mathbf{x}, \Omega) = \psi_{\text{inc}}(\mathbf{x}, \Omega) - \frac{1}{4\pi} \bar{\varphi}(\mathbf{x}), \quad \mathbf{x} \in \partial \mathcal{D} \text{ and } \Omega \cdot \mathbf{n} < 0. \quad (7b)$$

How do we choose the arbitrary function $\bar{\varphi}$ in the transformed problem defined by eqs. (7a) and (7b)? We need $(\varphi - \bar{\varphi}) \sim 1/\sigma_t$ in order to fix the noise issue because of how the variance depends on the total source function, which we will show in section 5. A simple choice would be $\bar{\varphi} = \varphi$. We could make this choice, and then account for the $\nabla \varphi$ source term by sourcing MC particles on the material interfaces. In a mesh where material interfaces coincide with interior mesh faces, and each mesh element contains a different material, this would require sourcing MC particles on every interior mesh face.

Our approach avoids interior face sources by applying finite element differentiation to compute a volumetric source that we sample instead of face sources. We choose $\bar{\varphi} \neq \varphi$ in order to avoid computing $\nabla \varphi$, which would introduce a well-posedness issue. This is because the solution of the transport equation is generally not differentiable, and so approximating the derivative of the transport solution will not give convergence no matter what numerical scheme is used. The formulation itself is not well-posed. Thus, we choose $\bar{\varphi}$ such that it satisfies the following conditions,

1. $\nabla \bar{\varphi}$ is well-defined, and
2. $(\varphi - \bar{\varphi}) \sim 1/\sigma_t$.

Numerical evidence suggests that the first condition is necessary for our hybrid method to converge to the correct solution in portions of phase space where the gradient of the transport solution may not be well-defined, such as on material interfaces. The second condition fixes the noise issue because $\text{Var}[\cdot] \sim \varepsilon$ instead of $1/\varepsilon$. Choosing $\bar{\varphi} = \varphi$ would satisfy the second condition, but not the first. We satisfy both conditions by solving a transient heat conduction equation for $\bar{\varphi}$,

$$\frac{\partial \bar{\varphi}}{\partial t} = \nabla \cdot \frac{1}{\sigma_t} \nabla \bar{\varphi}, \quad \mathbf{x} \in \mathcal{D}. \quad (8a)$$

Equation (8a) satisfies the first condition because it tends to produce differentiable solutions. This is true even when the initial condition is not differentiable, as in our case, because we use φ as the initial data:

$$\bar{\varphi}(\mathbf{x}, 0) = \varphi(\mathbf{x}), \quad \mathbf{x} \in \mathcal{D}. \quad (8b)$$

We satisfy the second condition by employing the following three techniques: i) choosing $1/\sigma_t$ for the conduction coefficient in eq. (8a), ii) choosing a Dirichlet boundary condition with φ as the Dirichlet data,

$$\bar{\varphi}(\mathbf{x}, t) = \varphi(\mathbf{x}), \quad \mathbf{x} \in \partial\mathcal{D}, \quad (8c)$$

and iii) computing a single timestep of backward Euler where the timestep size is subject to the limitation,

$$\frac{\Delta t}{h^2} \frac{1}{\max_{\mathbf{x}} \sigma_t} \ll 1. \quad (9)$$

We emphasize that the timestep in eq. (8a) is a *pseudo-time* used only to construct the auxiliary field $\bar{\varphi}$ in the source-transformed transport eq. (7a). It is not related to any physical time in the steady-state Boltzmann problem. Equation (9) is a *smallness* condition that limits the amount of smoothing in one pseudo-time step so that $\bar{\varphi}$ remains close to φ and $(\varphi - \bar{\varphi}) \sim 1/\sigma_t$, which is required for the transformed total source to remain bounded in the thick diffusion limit, as described in section 5. Other smallness conditions may work just as well as, or even better than, eq. (9).

Figure 2 shows the resulting HSM algorithm. The second moment system in the HSM algorithm remains unchanged from the SMM algorithm (see fig. 1). However, the transport equations differ, and the HSM algorithm has an additional transient heat conduction solve. We use Monte Carlo to solve the transport equation and a deterministic method to solve the moment system and the conduction equation. Monte Carlo avoids the ray effects which can appear in deterministic methods, such as discrete ordinates (S_N), due to under-resolution in angle. The moment system and the conduction equation do not depend on angle.

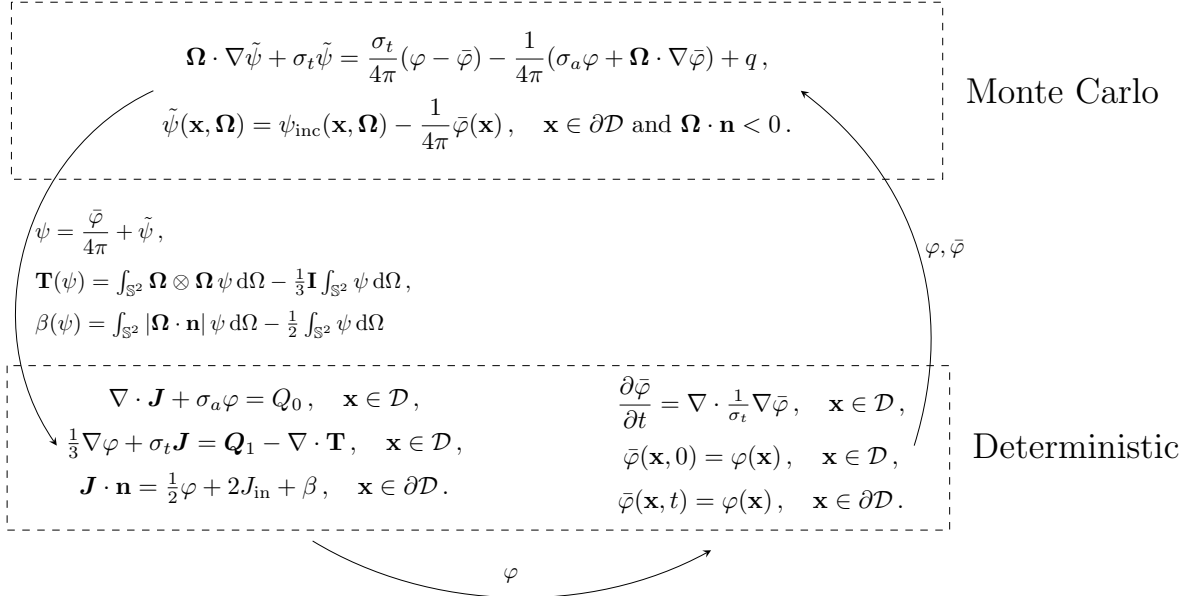


Figure 2: Hybrid second moment (HSM) algorithm.

The deterministic component of the hybrid method, which inverts the moment system, may produce non-realizable iterates $(\mathbf{J}^{(i)}, \varphi^{(i)})$, where i denotes the iteration index, though we have not observed this in practice. The hybrid algorithm remains well-defined even if realizability is not maintained at every

step, because the Monte Carlo component can still proceed with non-realizable iterates. This is enabled by representing signed quantities using negative weights. Importantly, the Monte Carlo solve requires negative weights to represent the source terms in eqs. (7a) and (7b), independent of whether the current iterates are realizable. This requirement arises from the source transformation variance reduction technique used to derive eqs. (7a) and (7b), which introduces unavoidable negative sources into the Monte Carlo calculation.

Monte Carlo particles sampled from sources that include negativities can have negative weights. Handling negative weights requires modifications to standard variance reduction techniques. For example, the logic that switches a particle's absorption mode from implicit capture to analog absorption when its weight falls below 1% of its initial value must be modified to use the absolute value of the weight, otherwise the switch would never occur. Under implicit capture, absorption of a negative weight particle increases its weight (that is, makes it less negative). The common feature shared by negative and positive weight particles is therefore not whether their weights increase or decrease under implicit capture, but that their weights decrease in absolute value. Another example arises when deciding how many particles to sample from each source. Instead of simply integrating the source function, one should compute its L_1 norm. Minimizing dispersion in the particle weight distribution then requires sampling relatively more particles from sources with larger L_1 norms.

Algorithm 1 shows pseudocode for the HSM algorithm, which uses Monte Carlo (MC) and the finite element method (FEM) to compute intermediate solutions, and where $\hat{\phi}$ is a MC estimator of the angle integrated intensity,

$$\phi = \int_{\mathbb{S}^2} \psi \, d\Omega, \quad (10)$$

and $\hat{\mathbf{T}}$ and $\hat{\beta}$ are MC estimators of $\mathbf{T}$ and β , respectively. The first MC transport solve uses the fixed source functions q and ψ_{inc} , whereas the subsequent MC transport solves use $\varphi^{(i)}$ and $\bar{\varphi}^{(i)}$, which are necessary to form the variable source functions. This allows us to use all of the particle memory to sample the fixed sources before iterating, then re-use all of the particle memory to sample the variable sources during the iteration. The iteration converges when the relative difference of successive iterates falls below a user-provided threshold η ,

$$\max_j \left(\frac{|\hat{\phi}_j^{(i-1)} - \hat{\phi}_j^{(i)}|}{\hat{\phi}_j^{(i-1)}} \right) < \eta, \quad j = 1, \dots, |\mathcal{T}|, \quad (11)$$

where $|\mathcal{T}|$ is the number of elements in the mesh.

Algorithm 1 Hybrid Second Moment

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1:  $\hat{\phi}^{(0)}, \hat{\mathbf{T}}^{(0)}, \hat{\beta}^{(0)} \leftarrow$  Solve eq. (7a) subject to eq. (7b) with  $(\varphi, \bar{\varphi}) = 0$  and  $(q, \psi_{\text{inc}}) \neq 0$  using MC
2:  $i \leftarrow 1$ 
3: while not converged( $\hat{\phi}^{(i-1)}, \hat{\phi}^{(i)}$ ) do
4:    $\varphi^{(i)} \leftarrow$  Solve eqs. (2a) and (2b) subject to eq. (2c) with data  $(Q_0, \mathbf{Q}_1, \hat{\mathbf{T}}^{(i-1)}, \hat{\beta}^{(i-1)})$  using FEM
5:    $\bar{\varphi}^{(i)} \leftarrow$  Solve eq. (8a) subject to eqs. (8b), (8c) and (9) with data  $\varphi^{(i)}$  using FEM
6:    $\hat{\phi}_{\text{temp}}, \hat{\mathbf{T}}_{\text{temp}}, \hat{\beta}_{\text{temp}} \leftarrow$  Solve eq. (7a) subject to eq. (7b) with  $(\varphi^{(i)}, \bar{\varphi}^{(i)})$  and  $(q, \psi_{\text{inc}}) = 0$  using MC
7:    $\hat{\phi}^{(i)} \leftarrow \hat{\phi}^{(0)} + \hat{\phi}_{\text{temp}}$ 
8:    $\hat{\mathbf{T}}^{(i)} \leftarrow \hat{\mathbf{T}}^{(0)} + \hat{\mathbf{T}}_{\text{temp}}$ 
9:    $\hat{\beta}^{(i)} \leftarrow \hat{\beta}^{(0)} + \hat{\beta}_{\text{temp}}$ 
10:   $i \leftarrow i + 1$ 
11: return  $\hat{\phi}^{(i)}$ 

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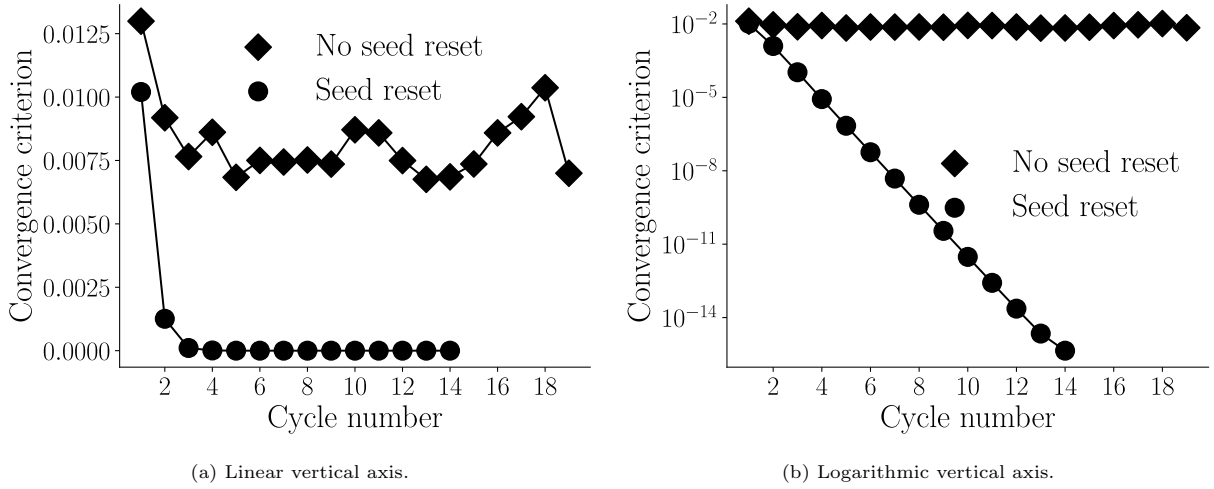


Figure 3: Resetting the pseudo-random number generator seed allows the HSM iteration to converge to machine precision for the convergence criterion specified in eq. (11). The information content of each particle history would be greater if the seed were not reset. However, this approach would require a more sophisticated convergence criterion which depends on more than just the current and previous iterates. For example, neutronics eigenvalue iteration algorithms typically do not reset the seed and often compute a moving average of iterates to assess convergence.

In algorithm 1, we recompute the particle weights every cycle, but the other parameters of the particles, as well as their histories, remain unchanged. We achieve this by resetting the pseudo-random number generator seed every cycle of the HSM iteration (see fig. 3). Finally, recall that the HSM method for solving eqs. (1a) and (1b) does not include scattering events. Thus, the histories of the particles in the MC transport solves in algorithm 1 do not include scattering events. The histories consist of a sequence of zero or more facet crossings terminated with absorption or leakage. The combination of having no scattering events, which allows each cycle to be computed quickly, and robust convergence, which ensures that only a few iterations—ten or fewer in all the calculations presented here—are required for convergence, makes HSM an effective algorithm for linear transport.

3. Deterministic Component of HSM

In this section, we derive the second moment system of eqs. (2a) and (2b) and the boundary condition eq. (2c). We use a finite element method to solve the second moment system, as well as the transient heat conduction eq. (8a), which we describe after presenting the weak form for each.

3.1. Derivation of the Second Moment System

We derive the second moment system of eqs. (2a) and (2b) and the boundary condition eq. (2c) following the procedure in Olivier and Haut [27]. Define the zeroth, first, and second angular moments of the radiation intensity ψ ,

$$\phi(\mathbf{x}) = \int_{\mathbb{S}^2} \psi(\mathbf{x}, \boldsymbol{\Omega}) d\boldsymbol{\Omega}, \quad (12)$$

$$\mathbf{J}(\mathbf{x}) = \int_{\mathbb{S}^2} \boldsymbol{\Omega} \psi(\mathbf{x}, \boldsymbol{\Omega}) d\boldsymbol{\Omega}, \quad (13)$$

$$\mathbf{P}(\mathbf{x}) = \int_{\mathbb{S}^2} \boldsymbol{\Omega} \otimes \boldsymbol{\Omega} \psi(\mathbf{x}, \boldsymbol{\Omega}) d\boldsymbol{\Omega}. \quad (14)$$

Define the zeroth and first angular moments of the arbitrary source function q ,

$$Q_0(\mathbf{x}) = \int_{\mathbb{S}^2} q(\mathbf{x}, \boldsymbol{\Omega}) d\boldsymbol{\Omega}, \quad (15)$$

$$\mathbf{Q}_1(\mathbf{x}) = \int_{\mathbb{S}^2} \boldsymbol{\Omega} q(\mathbf{x}, \boldsymbol{\Omega}) d\boldsymbol{\Omega}. \quad (16)$$

Taking the zeroth and first angular moments of eq. (1a) gives,

$$\nabla \cdot \mathbf{J} + \sigma_a \phi = Q_0, \quad (17a)$$

$$\nabla \cdot \mathbf{P} + \sigma_t \mathbf{J} = \mathbf{Q}_1. \quad (17b)$$

The system of eqs. (17a) and (17b) is unclosed because it has ten unknowns and only four equations when three spatial dimensions are considered. In 3D, the independent variables ϕ , $\mathbf{J}$, and $\mathbf{P}$ consist of 1+3+6=10 unknowns because they are a scalar, vector, and symmetric tensor, respectively; eq. (17a) is one equation, and eq. (17b) is three. The exact closure which defines the Second Moment Method (SMM) is,

$$\mathbf{P} = \mathbf{T} + \frac{1}{3} \mathbf{I} \phi, \quad (18)$$

where $\mathbf{T}$ is defined in eq. (4) and $\mathbf{I}$ is the identity matrix.

We derive a boundary condition by defining partial currents,

$$J_n^\pm = \int_{\boldsymbol{\Omega} \cdot \mathbf{n} \gtrless 0} \boldsymbol{\Omega} \cdot \mathbf{n} \psi d\boldsymbol{\Omega}, \quad (19)$$

and performing algebraic manipulation,

$$\begin{aligned} \mathbf{J} \cdot \mathbf{n} &= J_n^- + J_n^+ \\ &= 2J_n^- + (J_n^+ - J_n^-) \\ &= 2J_n^- + \int_{\mathbb{S}^2} |\boldsymbol{\Omega} \cdot \mathbf{n}| \psi d\boldsymbol{\Omega}. \end{aligned} \quad (20)$$

Defining the quantity,

$$B(\psi) = \int_{\mathbb{S}^2} |\boldsymbol{\Omega} \cdot \mathbf{n}| \psi d\boldsymbol{\Omega}, \quad (21)$$

and using J_{in} as defined in eq. (3) allows us to write the unclosed boundary condition,

$$\mathbf{J} \cdot \mathbf{n} = B(\psi) + 2J_{\text{in}}. \quad (22)$$

The exact closure for eq. (22) which defines the SMM is,

$$B = \beta + \frac{1}{2} \phi, \quad (23)$$

where β is defined in eq. (5). Substituting the exact SMM closure eq. (18) into eq. (17b), and the exact SMM closure eq. (23) into eq. (22) gives the second moment system of eqs. (2a) and (2b) and the boundary condition eq. (2c). Note that we switched from using ϕ in eqs. (17a), (18) and (23) to φ in eqs. (2a) to (2c) to emphasize that, even though the second moment system is an equivalent reformulation of the transport equation, φ can differ from the ϕ defined in eq. (12) after discretization, which is the case for our HSM method.

3.2. Mixed Finite Element Method for Solving the Second Moment System

The key consideration when solving the second moment system in our hybrid method is to avoid amplifying the noise in the SMM data, $\mathbf{T}$ and β . Computing $\mathbf{T}$ and β using Monte Carlo introduces noise, and differentiating a noisy quantity tends to amplify the noise. This suggests that we should solve the first order form of the system. Additionally, numerical evidence suggests that we should solve a weak form of eq. (2b) so that we can replace the strong derivative on $\mathbf{T}$ with a weak derivative. We use finite elements

to discretize space, and integration by parts to move the derivative from $\mathbf{T}$ to the test function. Integration by parts generates interior face integrals that we compute without interior face estimators. Instead, we use finite element interpolation of a volume estimator of $\mathbf{T}$ to compute $\mathbf{T}$ on the interior faces. Other approaches might also avoid noise amplification, and some may do so more efficiently; however, ours is a sufficient approach to avoid it.

The first order form is a mixed problem, which requires simultaneously solving for two quantities that belong to different finite element spaces. The FEM weak form for the mixed problem is: find $(\varphi, \mathbf{J}) \in Y_1 \times RT_0$ such that,

$$\int u \nabla \cdot \mathbf{J} \, d\mathbf{x} + \int \sigma_a u \varphi \, d\mathbf{x} = \int u Q_0 \, d\mathbf{x}, \quad \forall u \in Y_1, \quad (24a)$$

$$\begin{aligned} -\frac{1}{3} \int \nabla \cdot \mathbf{v} \varphi \, d\mathbf{x} + \int \sigma_t \mathbf{v} \cdot \mathbf{J} \, d\mathbf{x} + \frac{2}{3} \int_{\Gamma_b} (\mathbf{v} \cdot \mathbf{n})(\mathbf{J} \cdot \mathbf{n}) \, ds &= \int \mathbf{v} \cdot \mathbf{Q}_1 \, d\mathbf{x} - \int_{\Gamma_b} \mathbf{v} \cdot \mathbf{T} \mathbf{n} \, ds \\ + \frac{2}{3} \int_{\Gamma_b} (\mathbf{v} \cdot \mathbf{n})(2J_{\text{in}} + \beta) \, ds - \int_{\Gamma_0} \llbracket v \rrbracket \cdot \llbracket \mathbf{T} \mathbf{n} \rrbracket \, ds + \int \nabla_h \mathbf{v} : \mathbf{T} \, d\mathbf{x} &\quad \forall \mathbf{v} \in RT_0. \end{aligned} \quad (24b)$$

In eqs. (24a) and (24b), φ and $\mathbf{J}$ are the scalar and vector trial functions, respectively, u and $\mathbf{v}$ are the scalar and vector test functions, respectively, and Y_1 and RT_0 are the degree-1 Discontinuous Galerkin (DG) and degree-0 Raviart Thomas (RT) finite element spaces, respectively [28, 29]. Figure 4 shows the locations of the Y_1 and RT_0 degrees of freedom on the unit square.

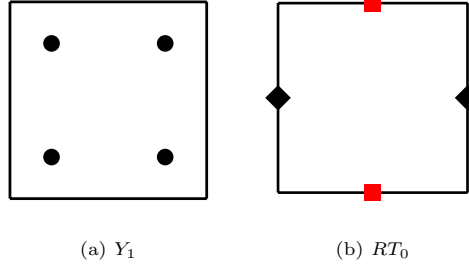


Figure 4: Location of the degrees of freedom for the (a) Y_1 and (b) RT_0 finite element spaces on a square element. In (b), the black diamonds and red squares denote degrees of freedom representing the x and y components, respectively.

The “broken” gradient ∇_h , the jump operator $\llbracket \cdot \rrbracket$, and the average operator $\{\!\{ \cdot \}\!\}$ are defined as,

$$(\nabla_h u)|_K = \nabla(u|_K) \quad \forall K \in \mathcal{T}, \quad (25)$$

$$\llbracket u \rrbracket = \begin{cases} u_1 - u_2 & \mathcal{F} \in \Gamma_0 \\ u & \mathcal{F} \in \Gamma_b \end{cases}, \quad (26)$$

$$\{\!\{ u \}\!\} = \begin{cases} \frac{u_1 + u_2}{2} & \mathcal{F} \in \Gamma_0 \\ u & \mathcal{F} \in \Gamma_b \end{cases}, \quad (27)$$

where K is an element in the mesh, $\mathcal{T}$ is the set of all mesh elements, $\mathcal{F}$ is a mesh element face, Γ_0 is the set of unique faces on the interior of the mesh, Γ_b is the set of unique faces on the boundary of the mesh, u_1 is the value of u on one side of the face, and u_2 is the value of u on the other side.

To derive eq. (24a), let $u \in Y_1$, multiply eq. (2a) by u , then integrate over $\mathcal{D}$. Deriving eq. (24b) requires several steps and employs integration by parts rules created using the following vector calculus identities:

- Product rule for divergence of a scalar (a) times a vector ($\mathbf{F}$):

$$\nabla \cdot (a\mathbf{F}) = \nabla a \cdot \mathbf{F} + a \nabla \cdot \mathbf{F}, \quad (28)$$

- Divergence theorem:

$$\int_K \nabla \cdot \mathbf{F} \, d\mathbf{x} = \int_{\partial K} \mathbf{F} \cdot \mathbf{n} \, ds, \quad (29)$$

- Product rule for divergence of a vector ($\mathbf{v}$) dotted with a tensor ($\mathbf{T}$):

$$\nabla \cdot (\mathbf{v} \cdot \mathbf{T}) = \mathbf{v} \cdot (\nabla \cdot \mathbf{T}) + \mathbf{T} : \nabla \mathbf{v}, \quad (30)$$

- Double dot product involving vectors ($\mathbf{v}$, $\mathbf{n}$) and a tensor ($\mathbf{T}$):

$$(\mathbf{v} \cdot \mathbf{T}) \cdot \mathbf{n} = \mathbf{v} \cdot (\mathbf{T} \mathbf{n}). \quad (31)$$

Combining eq. (28) with eq. (29) gives an integration by parts rule,

$$\int \nabla a \cdot \mathbf{F} \, d\mathbf{x} = - \int a \nabla \cdot \mathbf{F} \, d\mathbf{x} + \int_{\partial K} a (\mathbf{F} \cdot \mathbf{n}) \, ds. \quad (32)$$

Combining eqs. (28) to (30) gives another integration by parts rule,

$$\int \mathbf{v} \cdot \nabla \cdot \mathbf{T} \, d\mathbf{x} = - \int \mathbf{T} : \nabla \mathbf{v} \, d\mathbf{x} + \int_{\partial K} \mathbf{v} \cdot \mathbf{T} \mathbf{n} \, ds. \quad (33)$$

The three steps for deriving eq. (24b) are as follows:

Step 1) Integration over element K

Integrating eq. (2b) over an arbitrary finite element K and applying the integration by parts rules gives,

$$\begin{aligned} -\frac{1}{3} \int_K \nabla \cdot \mathbf{v} \, \varphi \, d\mathbf{x} + \frac{1}{3} \int_{\partial K} \varphi (\mathbf{v} \cdot \mathbf{n}) \, ds + \int_K \sigma_t \mathbf{v} \cdot \mathbf{J} \, d\mathbf{x} \\ = \int_K \mathbf{v} \cdot \mathbf{Q}_1 \, d\mathbf{x} - \int_{\partial K} \mathbf{v} \cdot \mathbf{T} \mathbf{n} \, ds + \int_K \nabla \mathbf{v} : \mathbf{T} \, d\mathbf{x} \quad \forall \mathbf{v} \in RT_0. \end{aligned} \quad (34)$$

Step 2) Sum over all elements

Summing eq. (34) over all elements $K \in \mathcal{T}$ gives,

$$\begin{aligned} -\frac{1}{3} \int \nabla \cdot \mathbf{v} \, \varphi \, d\mathbf{x} + \frac{1}{3} \int_{\partial \mathcal{D}} \varphi (\mathbf{v} \cdot \mathbf{n}) \, ds + \int \sigma_t \mathbf{v} \cdot \mathbf{J} \, d\mathbf{x} \\ = \int \mathbf{v} \cdot \mathbf{Q}_1 \, d\mathbf{x} - \int_{\partial \mathcal{D}} \mathbf{v} \cdot \mathbf{T} \mathbf{n} \, ds - \int_{\Gamma_0} \llbracket v \rrbracket \cdot \llbracket \mathbf{T} \mathbf{n} \rrbracket \, ds + \int \nabla_h \mathbf{v} : \mathbf{T} \, d\mathbf{x} \quad \forall \mathbf{v} \in RT_0. \end{aligned} \quad (35)$$

Step 3) Enforce boundary condition

“Solving” eq. (2c) for φ and substituting the result into the second term in eq. (35) gives the final result, eq. (24b).

An alternative derivation, as well as a hybridized Raviart-Thomas discretization of the second moment system, are provided by Olivier & Haut [27]. The hybridized discretization allows one to solve a smaller system for Lagrange multipliers, after which φ and $\mathbf{J}$ can be found through element-local back substitution.

Algorithm 2 shows pseudocode for the solve in line 4 of algorithm 1.

Algorithm 2 Mixed Finite Element Method Solve of Second Moment System

- 1: **function** Solve Second Moment System($Q_0, \mathbf{Q}_1, \mathbf{T}, \beta$)
 - 2: $A \leftarrow$ Assemble bilinear form for eqs. (24a) and (24b)
 - 3: $b \leftarrow$ Assemble linear form for eqs. (24a) and (24b) with data ($Q_0, \mathbf{Q}_1, \mathbf{T}, \beta$)
 - 4: $\varphi \leftarrow$ Solve linear system $A\varphi = b$
 - 5: **return** φ
-

3.3. Discontinuous Finite Element Method for Solving the Transient Heat Conduction Equation

As indicated in fig. 2, after we solve the second moment system for φ , we then solve a transient heat conduction equation for $\bar{\varphi}$. We use φ as the initial condition and the Dirichlet boundary condition in the standard symmetric interior penalty discontinuous Galerkin (SIPDG) method [30]. For clarity of presentation, we write the Dirichlet boundary data as $\bar{\varphi}_D$ instead of φ . The SIPDG weak form of eq. (8a) using linear basis functions is: find $\bar{\varphi} \in Y_1$ such that,

$$\begin{aligned} \frac{\partial}{\partial t} \int u \bar{\varphi} d\mathbf{x} + \int \nabla_h u \cdot \frac{1}{\sigma_t} \nabla_h \bar{\varphi} d\mathbf{x} - \int_{\Gamma} \llbracket u \rrbracket \left\{ \left\{ \frac{1}{\sigma_t} \nabla \bar{\varphi} \cdot \mathbf{n} \right\} \right\} ds - \int_{\Gamma} \left\{ \left\{ \frac{1}{\sigma_t} \nabla u \cdot \mathbf{n} \right\} \right\} \llbracket \bar{\varphi} \rrbracket ds \\ + \kappa \int_{\Gamma} \left\{ \left\{ \frac{1}{h} \frac{1}{\sigma_t} \right\} \right\} \llbracket u \rrbracket \llbracket \bar{\varphi} \rrbracket ds = - \int_{\Gamma_b} \frac{1}{\sigma_t} \nabla u \cdot \mathbf{n} \bar{\varphi}_D ds + \kappa \int_{\Gamma_b} \left\{ \left\{ \frac{1}{h} \frac{1}{\sigma_t} \right\} \right\} u \bar{\varphi}_D ds, \quad \forall u \in Y_1, \end{aligned} \quad (36)$$

where $\Gamma = \Gamma_0 \cup \Gamma_b$ is the set of unique faces in the mesh, κ is the penalty parameter, and h is the characteristic mesh element width,

$$h = \max_{K \in \mathcal{T}} \left(\int_K J d\xi \right)^{1/\dim}. \quad (37)$$

Equation (36) is the standard SIPDG weak form for the model elliptic problem with nonzero Dirichlet boundary conditions, and may be derived following the procedure outlined in Arnold et al. [31].

We discretize time using backward Euler. Let Φ contain the Y_1 degrees of freedom for $\bar{\varphi}$. The ordinary differential equation that arises after assembly of the weak form may be written,

$$M \frac{d\Phi}{dt} = -K\Phi + f, \quad (38)$$

where M, K , and f are the mass matrix, stiffness matrix, and load vector, respectively. Backward Euler is,

$$(M + \Delta t K)\Phi^{n+1} = M\Phi^n + \Delta t f, \quad (39)$$

where the superscript denotes the timestep index and Δt is the timestep size. We compute a single timestep with Δt equal to,

$$\Delta t = \alpha h^2 \left(\max_{\mathbf{x}} \sigma_t \right), \quad (40)$$

where we have introduced the timestep control parameter $\alpha \in (0, 1]$ into the timestep size eq. (9) to allow further control of the timestep size.

Algorithm 3 shows pseudocode for the solve in line 5 of algorithm 1. There is no time loop in algorithm 3 because we only take a single timestep.

4. Monte Carlo Component of HSM

We use Monte Carlo to solve the transport equation and compute estimators of the SMM data. Instead of using scattering events, we account for the scattering source by incorporating its contribution into the particle weights. While tracking, we accumulate estimators for the SMM data $\mathbf{T}$ and β . Specifically, we

Algorithm 3 Discontinuous Finite Element Method Solve of Transient Heat Conduction Equation

```

1: function Solve Transient Heat Conduction Equation( $\varphi$ )
2:    $\mathbf{M} \leftarrow$  Assemble mass matrix for eq. (36)
3:    $\mathbf{K} \leftarrow$  Assemble stiffness matrix for eq. (36)
4:    $\mathbf{f} \leftarrow$  Assemble linear form for eq. (36) with Dirichlet data  $\varphi$ , where  $\varphi$  is denoted by  $\bar{\varphi}_D$  for clarity
5:    $\bar{\varphi} \leftarrow$  Solve eq. (39) subject to eq. (40) with  $n = 0$  and initial data  $\Phi^0$ 
6:   return  $\bar{\varphi}$ 

```

compute $\hat{\mathbf{T}}$, which is an estimator for $\mathbf{T}$, using path length estimators in each element. Let $\text{vol}(K)$ be the volume of element K , d the length of a path traversed by a particle in K , and w the weight of the particle that traversed the path. Then,

$$\hat{\mathbf{T}} = \hat{\mathbf{P}} - \frac{1}{3} \mathbf{I} \hat{\phi}, \quad (41a)$$

$$\hat{\phi} = \frac{1}{\text{vol}(K)} \sum_i d_i w_i, \quad (41b)$$

$$\hat{\mathbf{P}} = \frac{1}{\text{vol}(K)} \sum_i \boldsymbol{\Omega}_i \otimes \boldsymbol{\Omega}_i d_i w_i, \quad (41c)$$

where the index i is associated with a particle path, so $\boldsymbol{\Omega}_i$ is the (entire) direction vector of the particle that traversed path i (*not* the i th component of the direction vector). Equations (41b) and (41c) are path length estimators for the zeroth and second angular moments of the intensity, respectively.

We compute $\hat{\beta}$, which is an estimator for β , using surface crossing estimators on each boundary face. Let $\text{area}(\mathcal{F})$ be the area of boundary face $\mathcal{F}$. Then,

$$\hat{\beta} = \hat{B} - \frac{1}{2} \hat{\phi}_s, \quad (42a)$$

$$\hat{B} = \frac{1}{\text{area}(\mathcal{F})} \sum_i w_i, \quad (42b)$$

$$\hat{\phi}_s = \frac{1}{\text{area}(\mathcal{F})} \sum_i \frac{w_i}{|\boldsymbol{\Omega}_i \cdot \mathbf{n}|}, \quad (42c)$$

Derivations of eqs. (41b), (41c), (42b) and (42c) can be found in Chapter 3 of Pozulp [32].

Every source is either a volume source in some subset of the domain, or a surface source on some subset of the domain boundary. Below, we describe how we compute weights for volume source particles and for boundary source particles. Consider the arbitrary source function $q = q(\mathbf{x}, \boldsymbol{\Omega})$, which is a volume source. Let $U(\cdot)$ be a uniformly-distributed random variate on some domain. For example, $U(a, b)$ is uniform on (a, b) . The weights that we assign to the particles sourced from the volume source q are,

$$w_i = \frac{4\pi \text{vol}(\mathcal{D})}{N} q(\mathbf{x}_i, \boldsymbol{\Omega}_i), \quad \mathbf{x}_i \leftarrow U(\mathcal{D}), \quad \boldsymbol{\Omega}_i \leftarrow U(\mathbb{S}^2), \quad (43)$$

where N is the number of particles sourced from q and $\leftarrow$ denotes the generation of a random variate. The weights that we assign to the particles sourced from the inflow boundary source $\psi_{\text{inc}}(\mathbf{x}, \boldsymbol{\Omega})$ are,

$$w_i = \frac{2\pi \text{area}(\partial\mathcal{D})}{M} |\boldsymbol{\Omega}_i \cdot \mathbf{n}| \psi_{\text{inc}}(\mathbf{x}_i, \boldsymbol{\Omega}_i), \quad \mathbf{x}_i \leftarrow U(\partial\mathcal{D}), \quad \boldsymbol{\Omega}_i \leftarrow U(\mathbb{S}_h^2), \quad (44)$$

where M is the number of particles sourced from ψ_{inc} , and $\mathbb{S}_h^2$ is the unit hemisphere at $\mathbf{x}_i$ defined by $\boldsymbol{\Omega}_i \cdot \mathbf{n} < 0$.

Algorithm 4 shows pseudocode for the solves in lines 1 and 6 of algorithm 1. There are no scattering events; the contribution of the scattering source is included in the particle weights.

Algorithm 4 Monte Carlo Solve of Linear Transport Equation

```

1: function Monte Carlo Transport Solve( $\varphi, \bar{\varphi}, q, \psi_{\text{inc}}$ )
2:    $\hat{\phi}, \hat{\mathbf{T}}, \hat{\beta} \leftarrow 0$ 
3:   Sample volume sources in eq. (7a) with data  $\varphi, \bar{\varphi}$ , and  $q$ 
4:   Sample boundary sources in eq. (7b) with data  $\bar{\varphi}$  and  $\psi_{\text{inc}}$ 
5:   Accumulate surface crossing estimators for boundary source particles
6:   while at least one particle has yet to absorb or leak do
7:      $d_f \leftarrow$  Compute distances to nearest facet along particle paths
8:      $d_a \leftarrow$  Sample distances to absorption
9:     Move particles to  $\mathbf{x}_{\text{new}} = \mathbf{x} + \min(d_f, d_a) \mathbf{\Omega}$ 
10:    Accumulate path length estimators
11:    Accumulate surface crossing estimators and leak particles for which  $\mathbf{x}_{\text{new}} \notin \mathcal{D}$ 
12:    Absorb particles for which  $d_a < d_f$ 
13:  return  $\hat{\phi}, \hat{\mathbf{T}}, \hat{\beta}$ 

```

5. Thick Diffusion Limit Analysis

In section 1, we noted that a Monte Carlo estimator of the solution to eq. (6) can have a variance that diverges in the thick diffusion limit (TDL), the asymptotic regime described by Larsen, Morel, and Miller [3]. We address this issue using a theorem from Pozulp and Haut [6]. We restate their theorem, along with their description of the TDL, because it provides a theoretical foundation for our numerical observation. Specifically, it explains why we need the source transformation variance reduction technique where we substitute $\psi = \bar{\varphi}/4\pi + \tilde{\psi}$ into the transport equation that we solve in the MC component of the HSM method.

5.1. The Variance Theorem

As described in [6], the TDL is an optically-thick, highly-scattering transport regime characterized by an asymptotic analysis parameter $\varepsilon \in (0, 1]$. Mathematically, the regime corresponds to a particular scaling of the transport problem data by powers of ε , followed by taking the limit $\varepsilon \rightarrow 0$. First, define new variables to represent the scaled problem data,

$$\sigma_t^\varepsilon = \sigma_t / \varepsilon, \quad (45a)$$

$$\sigma_a^\varepsilon = \sigma_a \varepsilon, \quad (45b)$$

$$q^\varepsilon = q \varepsilon, \quad (45c)$$

where $\sigma_a = \sigma_t - \sigma_s$ is the absorption opacity. The unscaled problem data are independent of ε and thus order unity,

$$\sigma_t \sim 1, \quad (46a)$$

$$\sigma_a \sim 1, \quad (46b)$$

$$q \sim 1. \quad (46c)$$

Therefore, the order dependence of the scaled problem data on ε is,

$$\sigma_t^\varepsilon \sim \varepsilon^{-1}, \quad (47a)$$

$$\sigma_a^\varepsilon \sim \varepsilon, \quad (47b)$$

$$q^\varepsilon \sim \varepsilon. \quad (47c)$$

Applying this scaling to the Boltzmann transport problem given by eq. (1a) with boundary condition eq. (1b) produces a new Boltzmann transport problem for ψ^ε ,

$$\mathbf{\Omega} \cdot \nabla \psi^\varepsilon + \sigma^\varepsilon \psi^\varepsilon = Q^\varepsilon, \quad (48a)$$

$$\psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) = \psi_{\text{inc}}^\varepsilon(\mathbf{x}, \mathbf{\Omega}), \quad \mathbf{x} \in \partial\mathcal{D} \text{ and } \mathbf{\Omega} \cdot \mathbf{n} < 0, \quad (48b)$$

where we used the symbol σ^ε in place of σ_t^ε for notational simplicity, and where Q^ε is defined as,

$$Q^\varepsilon = \frac{\sigma_t^\varepsilon - \sigma_a^\varepsilon}{4\pi} \int_{\mathbb{S}^2} \psi^\varepsilon d\Omega' + q^\varepsilon. \quad (48c)$$

A theorem relating the variance of Monte Carlo path length estimators for the new Boltzmann transport problem to the TDL parameter ε , along with a corollary, are stated in [6] and reproduced here. Proofs of the theorem and corollary may be found in [6].

Theorem 1. *Let $0 < \varepsilon \ll 1$ and let $\mathcal{D} \subset \mathbb{R}^{\text{dim}}$ ($\text{dim} = 1, 2, 3$) be the spatial domain, $\mathbb{S}^2$ the angular domain, $\partial\mathcal{D}$ the boundary of $\mathcal{D}$, and $\mathbb{S}_h^2$ the unit hemisphere of angles directed into $\mathcal{D}$ at a given point on $\partial\mathcal{D}$. Further, let $Q^\varepsilon : \mathcal{D} \times \mathbb{S}^2 \rightarrow \mathbb{R}$, $\psi_{\text{inc}}^\varepsilon : \partial\mathcal{D} \times \mathbb{S}_h^2 \rightarrow \mathbb{R}$, and $u^\varepsilon : \mathbb{S}^2 \rightarrow \mathbb{R}$ have leading terms in ε of $Q^\varepsilon \sim \varepsilon^{\alpha_Q}$, $\psi_{\text{inc}}^\varepsilon \sim \varepsilon^{\alpha_{\text{inc}}}$ and $u^\varepsilon \sim \varepsilon^{\alpha_u}$. Let $\sigma^\varepsilon = \frac{\sigma}{\varepsilon}$ with an otherwise ε -independent parameter $\sigma \in [0, \infty)$. Let ψ^ε solve,*

$$\mathbf{\Omega} \cdot \nabla \psi^\varepsilon + \sigma^\varepsilon \psi^\varepsilon = Q^\varepsilon, \quad (49)$$

with boundary data $\psi_{\text{inc}}^\varepsilon$ and $\mathbf{\Omega} \in \mathbb{S}^2$. Given $K \subset \mathcal{D}$, let $F = \frac{1}{\text{vol}(K)} \int_K \int_{\mathbb{S}^2} u^\varepsilon \psi^\varepsilon d\Omega d\mathbf{x}$ with $\mathbf{x} \in \mathcal{D}$ and let $\hat{F}$ be a Monte Carlo path length estimator of F with N interior and M boundary samples. Then there exists a function $G(\varepsilon)$ such that $\text{Var}[\hat{F}] \leq G(\varepsilon)$ where,

$$G(\varepsilon) \sim \varepsilon^{2\alpha_u} \left(\frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\text{inc}}+1}}{M} \right) \quad \text{as } \varepsilon \rightarrow 0. \quad (50)$$

Corollary 1. *Define the first three angular moments of the intensity by,*

$$\phi^\varepsilon = \int_{\mathbb{S}^2} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega, \quad (51)$$

$$\mathbf{J}^\varepsilon = \int_{\mathbb{S}^2} \mathbf{\Omega} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega, \quad (52)$$

$$\mathbf{P}^\varepsilon = \int_{\mathbb{S}^2} \mathbf{\Omega} \otimes \mathbf{\Omega} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega, \quad (53)$$

where $\otimes$ denotes the tensor (outer) product. Consider the averages of the moments over an arbitrary mesh element $K \subset \mathcal{D}$,

$$\phi_K^\varepsilon = \frac{1}{\text{vol}(K)} \int_K \phi^\varepsilon d\mathbf{x} = \frac{1}{\text{vol}(K)} \int_K \int_{\mathbb{S}^2} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega d\mathbf{x}, \quad (54)$$

$$\mathbf{J}_K^\varepsilon = \frac{1}{\text{vol}(K)} \int_K \mathbf{J}^\varepsilon d\mathbf{x} = \frac{1}{\text{vol}(K)} \int_K \int_{\mathbb{S}^2} \mathbf{\Omega} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega d\mathbf{x}, \quad (55)$$

$$\mathbf{P}_K^\varepsilon = \frac{1}{\text{vol}(K)} \int_K \mathbf{P}^\varepsilon d\mathbf{x} = \frac{1}{\text{vol}(K)} \int_K \int_{\mathbb{S}^2} \mathbf{\Omega} \otimes \mathbf{\Omega} \psi^\varepsilon(\mathbf{x}, \mathbf{\Omega}) d\Omega d\mathbf{x}. \quad (56)$$

Let $\hat{F}_0$, $\hat{F}_1$, and $\hat{F}_2$ denote the Monte Carlo path length estimators of ϕ_K^ε , $\mathbf{J}_K^\varepsilon$, and $\mathbf{P}_K^\varepsilon$, respectively, constructed as in Theorem 1 with the following choices of weighting function:

$$u_0(\mathbf{\Omega}) = 1, \quad (57)$$

$$\mathbf{u}_1(\mathbf{\Omega}) = \mathbf{\Omega}, \quad (58)$$

$$\mathbf{u}_2(\mathbf{\Omega}) = \mathbf{\Omega} \otimes \mathbf{\Omega}. \quad (59)$$

Then the following hold:

1. For $u_0 = 1$, the estimator $\hat{F}_0$ is a scalar path length estimator of the element average ϕ_K^ε , and its variance satisfies

$$\text{Var}[\hat{F}_0] \leq G_0(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\text{inc}}+1}}{M} \quad \text{as } \varepsilon \rightarrow 0. \quad (60)$$

2. For $\mathbf{u}_1 = \mathbf{\Omega}$, the estimator $\hat{F}_1$ is a vector-valued path length estimator of the element average $\mathbf{J}_K^\varepsilon$, and each component satisfies

$$\text{Var}[(\hat{F}_1)_\ell] \leq G_{1,\ell}(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\text{inc}}+1}}{M} \quad \text{as } \varepsilon \rightarrow 0, \quad \ell = 1, \dots, \dim. \quad (61)$$

3. For $\mathbf{u}_2 = \mathbf{\Omega} \otimes \mathbf{\Omega}$, the estimator $\hat{F}_2$ is a tensor-valued path length estimator of the element average $\mathbf{P}_K^\varepsilon$, and each component satisfies

$$\text{Var}[(\hat{F}_2)_{ij}] \leq G_{2,ij}(\varepsilon) \sim \frac{\varepsilon^{2\alpha_Q+1}}{N} + \frac{\varepsilon^{2\alpha_{\text{inc}}+1}}{M} \quad \text{as } \varepsilon \rightarrow 0, \quad i, j = 1, \dots, \dim. \quad (62)$$

5.2. Applying the Variance Theorem

We can use Theorem 1 and Corollary 1 to predict how HSM will perform in the TDL. Without the source transformation variance reduction technique, the Monte Carlo component of HSM would require computing Monte Carlo path length estimators associated with eq. (6) subject to the inflow boundary condition eq. (1b). The total source function Q for eq. (6) is the right-hand side of eq. (6),

$$Q = \frac{\sigma_s}{4\pi} \varphi + q. \quad (63)$$

Define $Q_{\text{HSM}}^\varepsilon$ to be Q from eq. (63) subject to the TDL scalings in eqs. (45a) to (45c). We can label the terms in $Q_{\text{HSM}}^\varepsilon$ with the leading order of ε for each term in the TDL. The terms in $Q_{\text{HSM}}^\varepsilon$ scale as,

$$Q_{\text{HSM}}^\varepsilon = \underbrace{\frac{\sigma_t/\varepsilon - \sigma_a\varepsilon}{4\pi} \varphi^\varepsilon}_{\varepsilon^{-1}} + \underbrace{q\varepsilon}_\varepsilon, \quad (64)$$

where we used $\varphi^\varepsilon \sim 1$. Thus, $Q_{\text{HSM}}^\varepsilon \sim \varepsilon^{-1}$ since $\varepsilon^{-1} \gg \varepsilon$ for $\varepsilon \in (0, 1] \rightarrow 0$, meaning that $\alpha_Q = -1$.

What are the exponents α_u and α_{inc} ? Determining α_u requires specifying a weighting function u and finding its TDL scaling u^ε . HSM requires computing the zeroth and second angular moments of the intensity, defined by eqs. (12) and (14), and estimated using the path length estimators in eqs. (41b) and (41c). Results 1. and 3. in Corollary 1 show that $\alpha_u = 0$ for both estimators. Determining α_{inc} requires a simplifying assumption: $\psi_{\text{inc}}^\varepsilon \sim 1$, so $\alpha_{\text{inc}} = 0$.

Substituting $\alpha_Q = -1$ and $\alpha_u = \alpha_{\text{inc}} = 0$ into Theorem 1 gives $\text{Var}[\cdot] \leq G(\varepsilon) \sim \varepsilon^{-1}$ for the path length estimators computed in the Monte Carlo component of the HSM method for the transport problem defined by eq. (6) subject to the boundary condition eq. (1b). The variance of Monte Carlo estimators used in the HSM method *without* the source transformation variance reduction technique would be unbounded in the TDL, making the HSM estimates practically unusable due to overwhelming statistical noise.

Now consider the transformed problem defined by eqs. (7a) and (7b). Define $Q_{\text{new}}^\varepsilon$ to be the total source function in eq. (7a). The terms in $Q_{\text{new}}^\varepsilon$ scale as,

$$Q_{\text{new}}^\varepsilon = \underbrace{\underbrace{\frac{\sigma_t/\varepsilon}{4\pi}}_{\varepsilon^{-1}} \underbrace{(\varphi^\varepsilon - \bar{\varphi}^\varepsilon)}_\varepsilon}_{1} - \underbrace{\frac{1}{4\pi} \left(\underbrace{\sigma_a\varepsilon\varphi^\varepsilon}_\varepsilon + \underbrace{\mathbf{\Omega} \cdot \nabla \bar{\varphi}^\varepsilon}_1 \right)}_1 + \underbrace{q\varepsilon}_\varepsilon, \quad (65)$$

where we used $\varphi^\varepsilon \sim 1$ and $\bar{\varphi}^\varepsilon \sim 1$. The terms in the new inflow boundary condition eq. (7b) scale as,

$$\tilde{\psi}(\mathbf{x}, \boldsymbol{\Omega}) = \underbrace{\psi_{\text{inc}}^\varepsilon(\mathbf{x}, \boldsymbol{\Omega})}_1 - \underbrace{\frac{1}{4\pi}\bar{\varphi}^\varepsilon(\mathbf{x})}_1, \quad (66)$$

where we have assumed once more that $\psi_{\text{inc}}^\varepsilon \sim 1$. As before, $\alpha_u = \alpha_{\text{inc}} = 0$, but now $\alpha_Q = 1$ instead of -1 . By Theorem 1, $\text{Var}[\cdot] \leq G(\varepsilon) \sim \varepsilon$ for the path length estimators computed in the Monte Carlo component of the HSM method for the transformed problem defined by eqs. (7a) and (7b). The variance limits to zero.

6. Numerical Results

We implemented an MPI-parallel Monte Carlo solver in the C programming language which distributes the simulation particles evenly amongst the MPI ranks. All ranks source and track particles. To implement the second moment system and the transient heat conduction equation solvers, we used the C++ programming language with the MPI-parallel finite element library MFEM [33] along with the MPI-parallel sparse iterative linear solvers library HYPRE [34] and the MPI-parallel sparse direct linear solver library SuperLU [35]. We used HYPRE’s algebraic multigrid preconditioner with the conjugate gradient solver to invert the coefficient matrix in the discretized transient heat conduction system defined by eq. (39). We used SuperLU to invert a Schur complement of one of the diagonal blocks of the second moment system block operator after computing the inverse of another block of element matrices using element-wise Gauss-Jordan elimination with partial pivoting. We used SuperLU because, in the largest problem we considered, inverting the second moment system accounted for less than 4% of the total calculation runtime. This allowed us the luxury of computing the solution to (a small multiple of) machine precision. However, SuperLU is not required; fast iterative linear solvers can be used to invert the second moment system and have been shown to be efficient on large second moment systems for various polynomial orders, as demonstrated by Olivier & Haut [27]. We set the penalty parameter to $\kappa = 4$ and the timestep control parameter to $\alpha = 0.001$. We source the same number of particles in the domain and on its boundary, so $N = M$ in eqs. (43) and (44).

The computational complexity and memory footprint of our implementation can be determined by considering the Monte Carlo and deterministic components individually. Each Monte Carlo transport solve costs $O(N_p \bar{n}_{\text{seg}})$, where $N_p = N + M$ is the number of particles, $\bar{n}_{\text{seg}}$ is the average number of segments per particle, and a sequence of “segments” constitute a particle history. The total Monte Carlo cost is $O((I+1)N_p \bar{n}_{\text{seg}})$, where I is the number of iterations required to converge the scattering source. Let N_e be the number of mesh elements. The deterministic moment system and transient heat conduction equation scale linearly in N_e for assembly and, with appropriate preconditioning, approximately linearly for solution if we use a parallel sparse iterative linear solver. However, we used SuperLU instead of an iterative solver because we found it to be sufficient for the problems we consider, so our linear solve time complexity is superlinear in N_e . Memory is $O(N_e)$ for deterministic operators and Monte Carlo estimators. Monte Carlo particle memory is $O(N_p)$.

We present numerical results for different linear transport problems modeled by eq. (1a) with the boundary condition eq. (1b). In section 6.1, we solve a problem with a single material to verify that our HSM method has a numerical error that agrees with our expectation for how the error should depend on the number of particles and the element width. In section 6.2, we solve another single-material problem, the purpose of which is to show how the HSM solution behaves as the optical thickness of the material increases. Finally, in section 6.3, we solve a proxy problem from radiative transfer that contains both optically-thick and optically-thin material, and show that our hybrid method runs orders of magnitude faster than Monte Carlo, without sacrificing accuracy. Our numerical results provide what we believe to be the first demonstration of a hybrid SMM in more than one spatial dimension, the first to consider more than one material, and the first to use variance reduction.

6.1. Method of Manufactured Solutions Problem

The Method of Manufactured Solutions (MMS) is a code verification technique in which one chooses an analytic solution, which is called the “MMS solution”. One then substitutes the MMS solution into the

partial differential equation to derive expressions for the problem data, which includes: sources, boundary conditions, and the initial condition [36, 37, 38]. We use MMS to verify that our HSM method has a numerical error of $O(h) + O(N^{-1/2})$, where h is the characteristic mesh length defined by eq. (37), and N is the number of MC particles. The first term is due to the h^{p+1} convergence of the mixed finite element discretization of the second moment system, where $p = 0$ in our case because we use Y_1 , RT_0 , and piecewise constant MC estimators for the SMM data. The second term is due to the $\sqrt{\text{Var}[\cdot]/N}$ uncertainty of the estimators computed in the Monte Carlo component of HSM. The hypothesized convergence rate of $O(h) + O(N^{-1/2})$ defines an error surface that decreases in height as one traverses simultaneously upward and rightward in fig. 5.

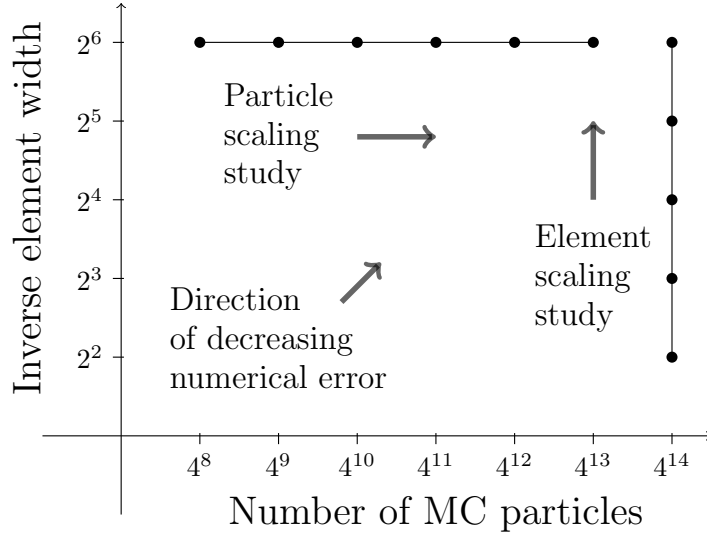


Figure 5: Calculation points on the HSM error surface.

By making h small and running multiple calculations with increasing N in the “particle scaling study”, we observe the $O(N^{-1/2})$ term in the hypothesized convergence rate, and by making N large and running multiple calculations with decreasing h in the “element scaling study”, we observe the $O(h)$ term. We let $\mathcal{D} = [0, 1]^2$, $\sigma_t = 2$, $\sigma_s = 1$, and we solve the problem specified by eqs. (1a) and (1b) for the MMS solution in Equation (89) in [39]. The MMS solution is,

$$\begin{aligned} \psi_{\text{MMS}}(\mathbf{x}, \boldsymbol{\Omega}) = & \frac{1}{4\pi} \left(\sin(\pi x) \sin(\pi y) + \boldsymbol{\Omega}_x \boldsymbol{\Omega}_y \sin(2\pi x) \sin(2\pi y) \right. \\ & \left. + \boldsymbol{\Omega}_x^2 \sin\left(\frac{5\pi}{2}x + \frac{\pi}{4}\right) \sin\left(\frac{5\pi}{2}y + \frac{\pi}{4}\right) + 0.5 \right). \end{aligned} \quad (67)$$

We substitute the MMS solution into the equations on the left of fig. 1, solve for q and ψ_{inc} , and use them in the MC component of our HSM solver. Similarly, we substitute the MMS solution into the equations on the right of fig. 1, solve for Q_0 , $\mathbf{Q}_1$, and J_{in} , and use them in the deterministic component of our HSM solver. The MMS source expressions for q , ψ_{inc} , Q_0 , $\mathbf{Q}_1$ and J_{in} can be found in Section 5.1.1 of [32].

Equation (67) cannot be computed with the radiation diffusion approximation because it is quadratically-anisotropic. Additionally, eq. (67) exercises the coupling of the MC component of HSM with the deterministic component of HSM because it has nonzero SMM data $\mathbf{T} \neq 0$ and $\beta \neq 0$. Finally, eq. (67) also exercises our implementation of the boundary conditions because it has nonzero inflow $\psi_{\text{inc}} \neq 0$, and so $J_{\text{in}} \neq 0$.

The solution that we seek with HSM is the angle integrated intensity, which we can express analytically

by integrating eq. (67) over the unit sphere to get,

$$\phi_{\text{MMS}}(\mathbf{x}) = \sin(\pi x) \sin(\pi y) + \frac{1}{3} \sin\left(\frac{5\pi}{2}x + \frac{\pi}{4}\right) \sin\left(\frac{5\pi}{2}y + \frac{\pi}{4}\right) + 0.5. \quad (68)$$

The HSM estimator approximates the average angle integrated intensity on each element. Averaging eq. (68) over an arbitrary rectangle gives the analytic expression to which we compare our HSM solution,

$$\bar{\phi}_{\text{MMS}}(x_1, x_2, y_1, y_2) = \frac{1}{(x_2 - x_1)(y_2 - y_1)} \int_{x_1}^{x_2} \int_{y_1}^{y_2} \phi_{\text{MMS}}(\mathbf{x}) \, dy \, dx. \quad (69)$$

Evaluating the integrals in eq. (69) gives the final analytic expression,

$$\begin{aligned} \bar{\phi}_{\text{MMS}}(x_1, x_2, y_1, y_2) = & \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left(\frac{1}{\pi^2} \left\{ \left[\cos(\pi x_2) - \cos(\pi x_1) \right] \left[\cos(\pi y_2) - \cos(\pi y_1) \right] \right\} \right. \\ & \left. + \frac{4}{75\pi^2} \left\{ \left[\cos\left(\frac{5\pi}{2}x_2 + \frac{\pi}{4}\right) - \cos\left(\frac{5\pi}{2}x_1 + \frac{\pi}{4}\right) \right] \left[\cos\left(\frac{5\pi}{2}y_2 + \frac{\pi}{4}\right) - \cos\left(\frac{5\pi}{2}y_1 + \frac{\pi}{4}\right) \right] \right\} \right). \end{aligned} \quad (70)$$

We use a computational mesh of equally-sized squares, so we simply evaluate eq. (70) using the x_1, x_2, y_1 , and y_2 values which define the left, right, bottom, and top edges of each square in our mesh. Figure 6 shows plots of ϕ_{MMS} and $\bar{\phi}_{\text{MMS}}$ on the calculation domain $\mathcal{D} = [0, 1]^2$.

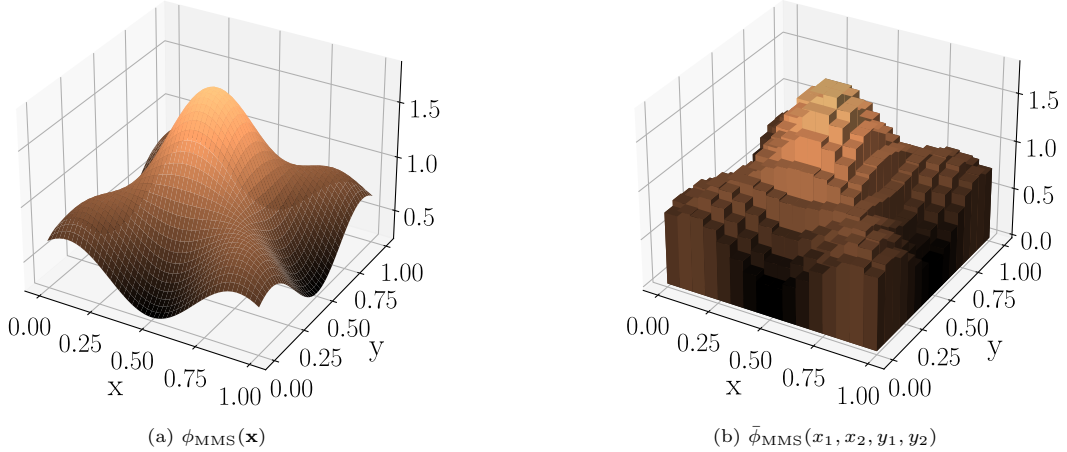


Figure 6: The MMS solution eq. (68) and its average eq. (70) on $\mathcal{D} = [0, 1]^2$.

We determine the error of our numerical solution with respect to eq. (70) in the element scaling study by computing the norm of the difference and multiplying by the characteristic mesh element width,

$$\text{Element scaling study error} = h \|\hat{\phi} - \bar{\phi}_{\text{MMS}}\|_2, \quad (71)$$

where h is the characteristic mesh length defined by eq. (37). In the particle scaling study, the mesh is fixed at 64×64 elements for all 6 calculations, so the normalization factor h in eq. (71) is replaced by unity,

$$\text{Particle scaling study error} = \|\hat{\phi} - \bar{\phi}_{\text{MMS}}\|_2. \quad (72)$$

Derivations of eqs. (71) and (72) may be found in Section 5.1 of [32].

The results in fig. 7 demonstrate that HSM converges under mesh refinement and MC sample augmentation, and that the rate of convergence with respect to the element width and number of MC particles

matches our hypothesis of $O(h) + O(N^{-1/2})$. The slight degradation in convergence at the highest mesh resolution, appearing as liftoff above the dashed curve at the bottom-left of fig. 7b, is expected behavior because we distribute a fixed number of MC particles across more and more elements under mesh refinement. We confirmed that this was the case by running fewer MC particles and observing earlier liftoff, as well as running more MC particles and observing delayed liftoff.

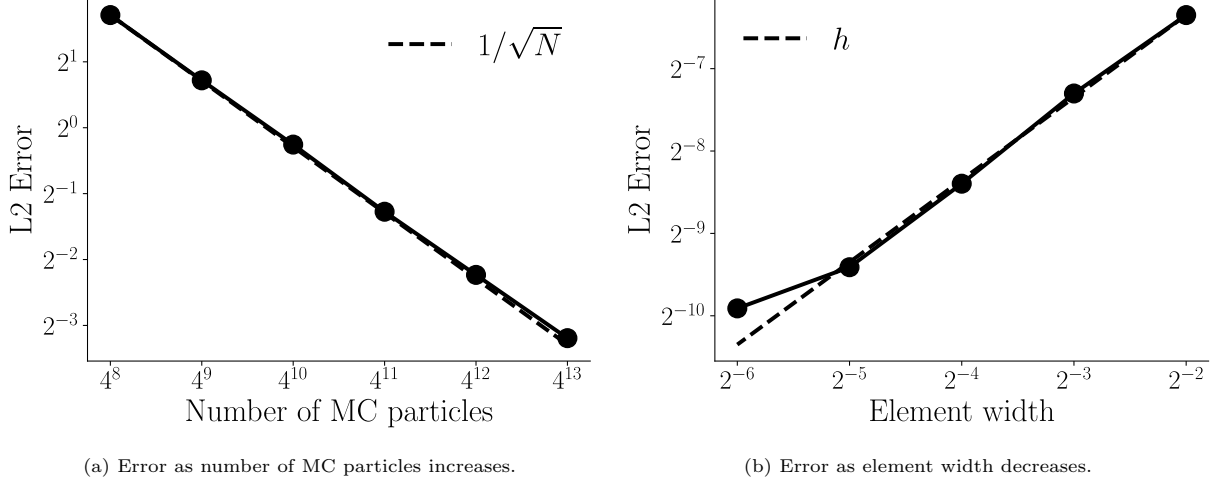


Figure 7: Error of HSM iterate $\hat{\phi}^{(i)}$ upon convergence.

6.2. Thick Diffusion Limit Problems

The thick diffusion limit, characterized by the parameter $\varepsilon \in (0, 1]$ when $\varepsilon \rightarrow 0$ for the problem data scalings defined in eqs. (45a) to (45c), is a challenging regime for transport methods. Unaccelerated source iteration calculations require $\sim 1/\varepsilon^2$ iterations to converge, and unaccelerated MC calculations require $\sim 1/\varepsilon^2$ runtime to finish. We show that the HSM runtime is independent of ε , and that our source transformation variance reduction technique provides the bounded estimator variance $\text{Var}[\cdot] \sim \varepsilon$, rather than $\text{Var}[\cdot] \sim 1/\varepsilon$.

Let $\mathcal{D} = [0, 1]^2$ as in section 6.1, but now define the problem data in eqs. (1a) and (1b) using the TDL scaling eqs. (45a) to (45c). Thus, unlike in our MMS calculation where $\sigma_t = 2$ and $\sigma_s = 1$, here we use $\sigma_t = 1/\varepsilon$ and $\sigma_s = 1/\varepsilon - \varepsilon$. Also, in our MMS calculation, q is a complicated expression derived by substituting the MMS solution into eq. (1a). Here, we use the very simple expression $q = \varepsilon$.

Consider four different ε values: 10^{-1} , 10^{-2} , 10^{-3} , and 10^{-4} . We run calculations for each value using 10 million HSM volume source particles, no boundary source particles, and a vacuum boundary condition. We compute the angle integrated intensity averaged over each element in an 8-by-8 mesh of squares of equal size. Our solution is a surface on the xy -plane. We present HSM solution lineouts. Figure 8 shows that our HSM solution appears to converge to the diffusion limit solution as $\varepsilon \rightarrow 0$ because the lineouts appear to approach an equilibrium solution in the sense that successive lineouts differ less for smaller ε .

What about runtime? The convergence depicted by fig. 8, which shows solution lineouts for four calculations using the HSM method, would be the same picture for unaccelerated Monte Carlo (UMC)¹. However, the runtimes of HSM and UMC in the TDL are very different. Figure 9 shows that the HSM runtime is constant with respect to the TDL scaling parameter, whereas the UMC runtime increases quadratically with the inverse of ε . HSM substantially outperforms UMC in terms of runtime as $\varepsilon \rightarrow 0$.

What about variance? A common figure of merit used with MC methods is the inverse of the product of the variance and the runtime,

$$\text{Figure of merit} = \frac{1}{\text{Var}[\cdot] T}, \quad (73)$$

¹We use the acronym UMC (instead of MC) to emphasize that the solution we compare with our HSM approach does not employ any of the acceleration techniques described in section 1, such as Fleck & Canfield's Random Walk, IMD, or DDMC.

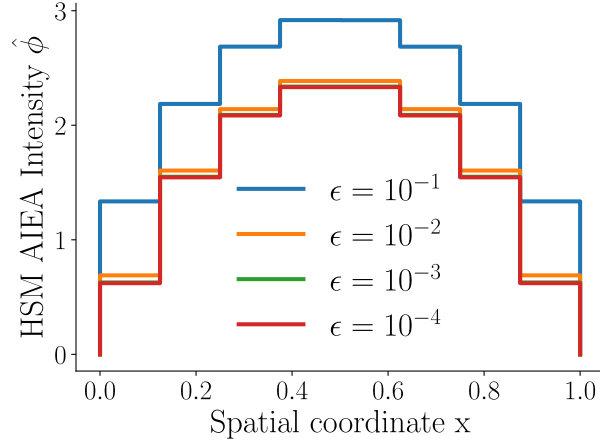


Figure 8: Lineouts at $y = 0.5$ of HSM angle integrated element averaged (AIEA) intensity for calculations using four different values of the TDL scaling parameter ϵ .

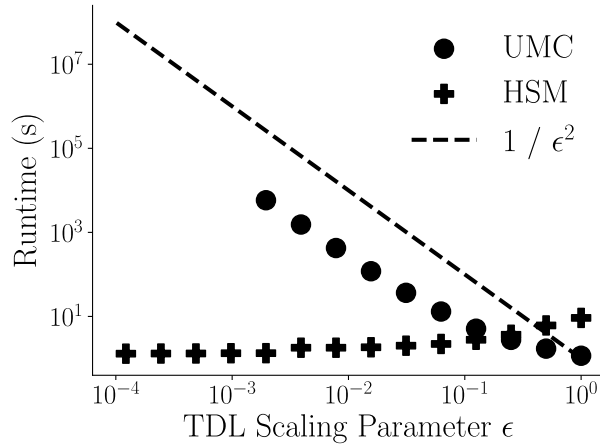


Figure 9: Runtime of unaccelerated Monte Carlo (UMC) and hybrid second moment (HSM) methods for calculations with different values of the TDL scaling parameter ϵ . The ϵ values are $2^0, 2^{-1}, 2^{-2}, \dots, 2^{-13}$.

where $\text{Var}[\cdot]$ is the variance of some estimator and T is the runtime of the calculation. Decreasing either $\text{Var}[\cdot]$ or T increases the figure of merit. Reading fig. 9 from right to left, we observe that $T_{\text{HSM}} < T_{\text{UMC}}$ after ε falls below a value somewhere between $\varepsilon = 1/4$ and $1/8$, and that $T_{\text{HSM}} \ll T_{\text{UMC}}$ beginning at about $\varepsilon = 1/32$, which is a result that holds for all smaller ε values. This factor of ε^2 decrease in the runtime would be useless if the variance of the HSM estimator increased by a factor of $1/\varepsilon^2$ relative to the UMC variance, because eq. (73) shows that the figure of merit would be left unchanged.

We calculate the variance of the HSM solution in the TDL by running 600 realizations of the same calculation on a single-element mesh with a different pseudo-random number generator seed for each calculation. Each calculation computes the angle integrated intensity averaged over the lone mesh element. The domain is $\mathcal{D} = [0, 1]^2$. We use 100 million HSM particles for each calculation. We consider five different optical thickness parameter values, 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} , and 10^{-5} . Thus, the total number of calculations is $600 \times 5 = 3000$. We estimate the variance as the sum of squares of the difference of the HSM angle integrated element averaged intensity estimator realizations with the mean estimate,

$$\text{Var}[\hat{\phi}] \approx \frac{1}{600} \sum_{i=1}^{600} (\hat{\phi}_i - \Phi)^2, \quad (74)$$

where Φ is the mean estimate defined by $\Phi = (1/600) \sum_{i=1}^{600} \hat{\phi}_i$. Figure 10 shows that the empirical variance of the HSM estimator $\hat{\phi}$ in the TDL before variance reduction is $\text{Var}[\cdot] \sim 1/\varepsilon$, which matches our expectation based on the fact that $\text{Var}[\hat{\phi}] \sim 1/\varepsilon$ for $Q^\varepsilon \sim 1/\varepsilon$, and after variance reduction is $\text{Var}[\hat{\phi}] \sim \varepsilon$, which matches our expectation based on the fact that $\text{Var}[\hat{\phi}] \sim \varepsilon$ when $Q^\varepsilon \sim 1$.

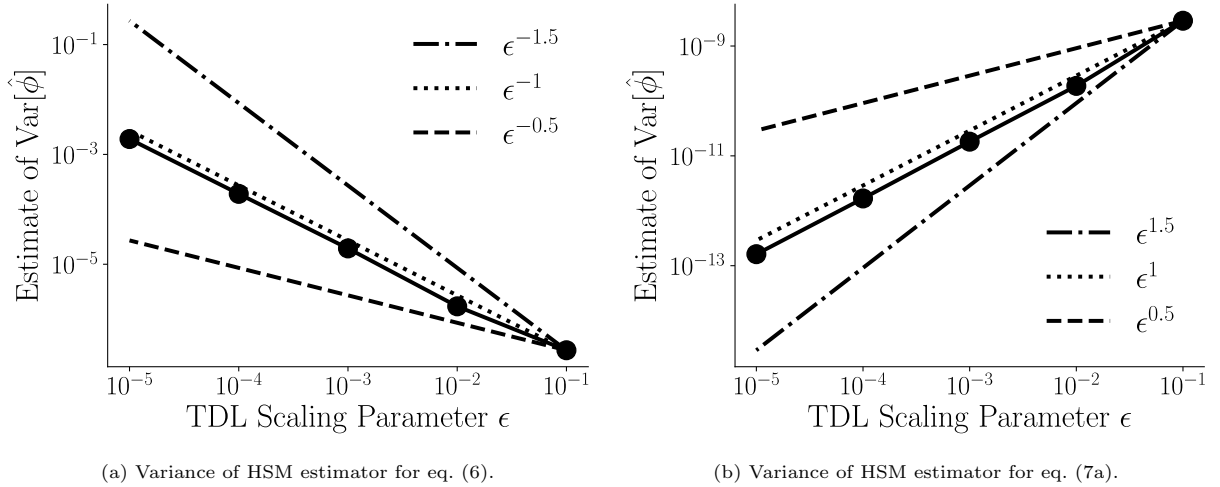


Figure 10: Hybrid Second Moment estimator variance before variance reduction (a) and after variance reduction (b).

6.3. Linearized Crooked Pipe Problem

We now consider a problem containing both optically-thick and optically-thin material. The Crooked Pipe is a thermal radiative transfer benchmark problem proposed by Graziani and LeBlanc [40]. The authors describe radiation flow down a pipe with a bend. In the Crooked Pipe problem description, a boundary source of radiation shines on one end of an optically-thin pipe surrounded by an optically-thick medium. Radiation flows preferentially through the pipe, around a bend in the pipe, and then leaves the problem through a vacuum boundary on the other end of the pipe.

Crooked Pipe is a time-dependent problem. The solution described in the problem statement is the temperature time-series at five fiducial points inside the pipe. We solve the Linearized Crooked Pipe, which is a steady-state version of Crooked Pipe defined in [39]. The linearization is equivalent to calculating one

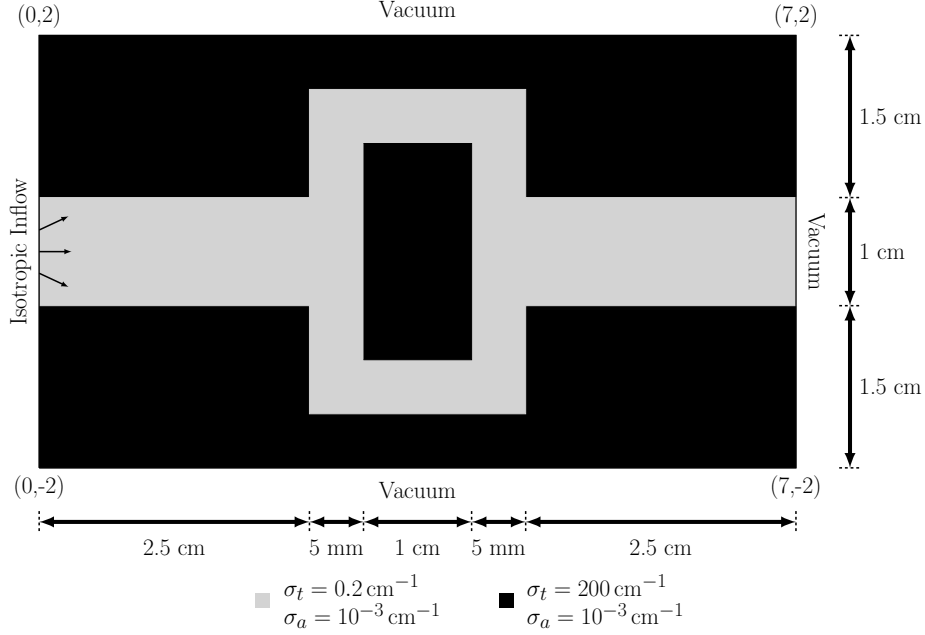


Figure 11: Linearized Crooked Pipe geometry, material data, and boundary conditions [39].

very large timestep of a backward Euler time integrator. The timestep is such that $c\Delta t = 10^3$, where c is the speed of light in a vacuum. The pipe is 1000 times optically thinner than the surrounding medium. Figure 11 depicts the geometry, material data, and boundary conditions for the problem. The boundary source is isotropic on the inflow hemisphere with magnitude,

$$\psi_{\text{inc}} = \begin{cases} \frac{1}{2\pi} & x = 0 \text{ and } -0.5 < y < 0.5, \\ 0 & \text{otherwise.} \end{cases} \quad (75)$$

Finally, there is an isotropic volume source with constant value $q = 10^{-7}$ everywhere in space, which emulates the material emission source term in the thermal radiative transfer equations.

Consider a mesh composed of squares of equal sizes. The minimal number of uniform squares required to tessellate the geometry such that all material interfaces are on element boundaries is $14 \times 8 = 112$. Figure 12a shows the aforementioned mesh elements, all of which contain material with single-valued properties, because each element contains either optically-thin pipe material, or optically-thick surrounding material, but not both. Figure 12b shows the mesh after splitting each element into four squares of equal size to create a new mesh with 28×16 elements. Repeating this refinement operation three more times results in a 224×128 mesh, with 28,672 square elements, which is the mesh that we used in our calculations. We set the convergence threshold η in the convergence criterion eq. (11) to 10^{-3} . We computed a piecewise constant angle integrated intensity using our $\hat{\phi}$ estimator of ϕ . Our estimator approximates the average of ϕ on each element, meaning that it is single-valued within a mesh element.

We ran three calculations. We computed HSM and UMC solutions using 16 million simulation particles. We also computed a radiation diffusion approximation solution. Figure 13 shows pseudocolor plots of the three solutions. Three comments regarding the results in fig. 13 are:

1. The HSM and UMC solutions agree to about 3% in the infinity norm.
2. The bright illumination of the front wall and the dark shadowing of the back wall are the most prominent in UMC, remain visible in HSM, and are nearly indiscernible in the diffusion approximation.

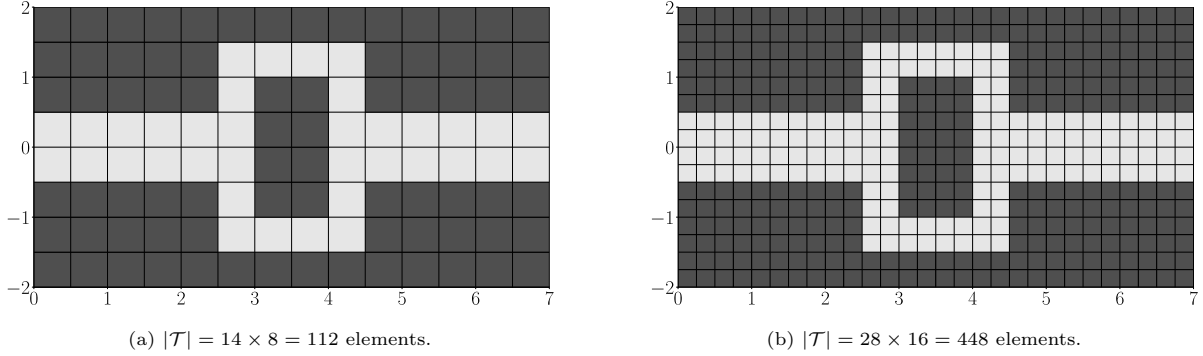


Figure 12: Minimal uniform square mesh of the Linearized Crooked Pipe geometry with single-valued material property elements (a), and the same mesh after splitting each element into four squares of equal size (b).

3. Similarly, the section of pipe nearest to the right boundary is darkest in UMC and brightest in the diffusion approximation, with HSM somewhere in between.

Without variance reduction, HSM requires about 500 times more simulation particles—8 billion instead of 16 million—to provide a solution with as little noise as UMC. While we do not present the noisy solution here, it can be found in the final figure of [6], which includes a pseudocolor plot of the noisy solution.

What about runtime? We ran the HSM and UMC calculations with 500 and 100 different pseudo-random number seeds, respectively, and found that the median runtime was about two orders of magnitude faster for HSM (which is why we ran five times more HSM calculations). The substantial runtime savings is due to the cost of the billions of scattering events in the UMC calculation, which we completely avoid by not having scattering events in the HSM algorithm.

What about variance? When calculating the variance for the collection of 500 HSM calculations and 100 UMC calculations, we get one value per mesh element. We desire a transport method with high global accuracy, so we choose to compare the maximum of the element variances. We found that the maximum variance is about one order of magnitude lower for UMC than for HSM. Thus, the figure of merit defined by eq. (73) is about one order of magnitude greater for HSM than UMC. Table 1 summarizes these results.

Figure 14 shows pseudocolor plots of the variance, revealing that the UMC variance is highest in the optically-thick material, while the HSM variance peaks at the inflow boundary. This suggests that we could reduce the HSM variance by sourcing more particles in the elements near the inflow boundary during the HSM iteration. In general, it would be better to sample the variable sources non-uniformly. This approach would provide a particle weight distribution with reduced dispersion, thereby decreasing the HSM variance. These results suggest that a bounded variance in the thick diffusion limit may come at the cost of a relatively higher variance in the optically thin portion of a mixed thick-thin problem, or higher overall variance in optically thin problems.

Finally, we collected a program counter sampling profile and observed that the HSM runtime is unevenly distributed between the deterministic and Monte Carlo components: 8% and 88%, respectively. The remaining 4% was spent in input/output, initialization, and “glue code” connecting the two components. We collected the profile using gperftools [41] while running HSM using 112 MPI ranks on one node of a system with two 56-core Intel Xeon Platinum 8480+ sockets and 256 GB of DRAM per node. The relatively high amount of runtime spent in the Monte Carlo component is consistent with the intuition that solving a transport equation using Monte Carlo is more expensive than using a deterministic method to invert a radiation diffusion operator.

7. Conclusions

In this work, we presented what we believe to be the first demonstration of a hybrid Second Moment Method for linear transport in more than one spatial dimension, the first to consider more than one material,

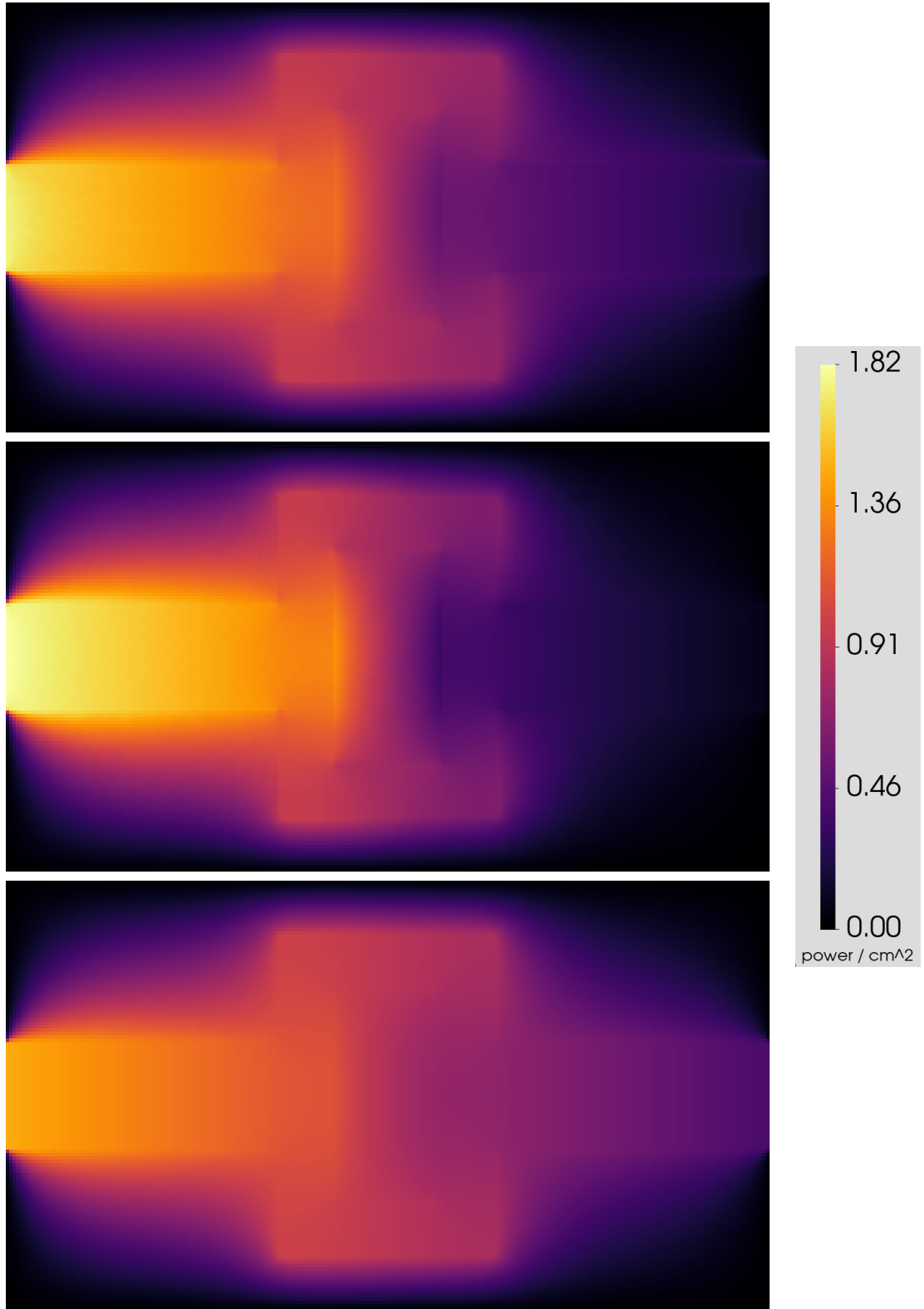
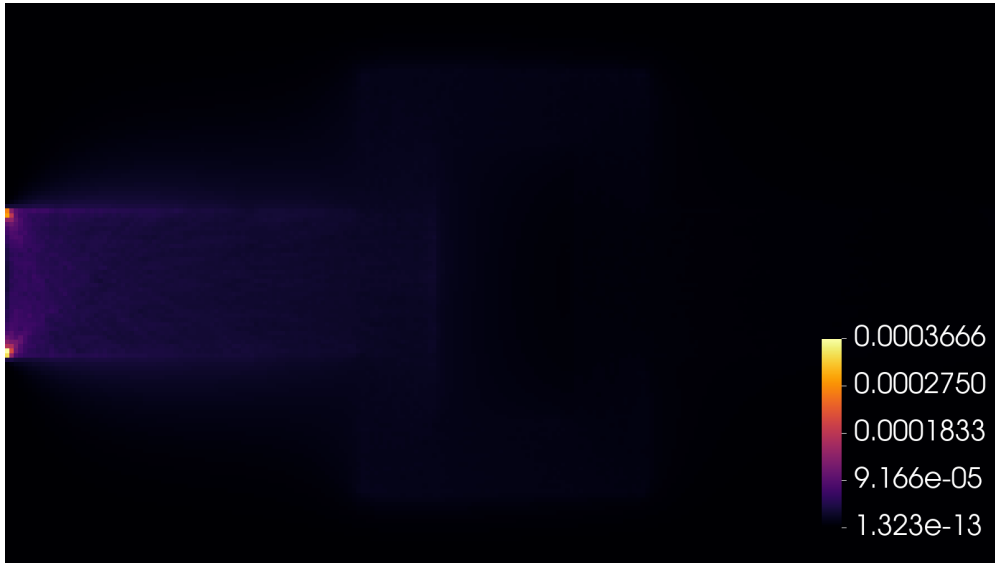


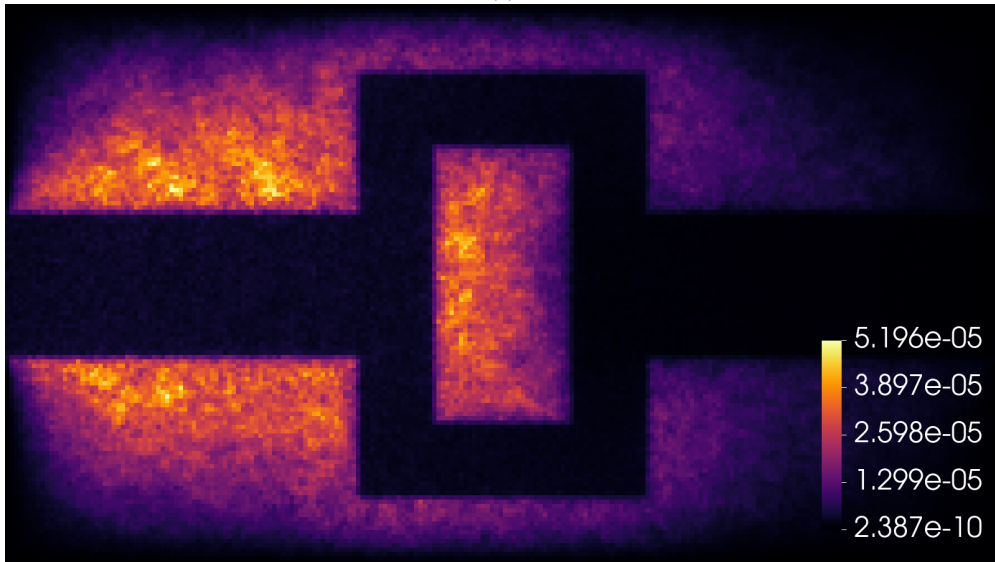
Figure 13: HSM (top), UMC (middle), and diffusion (bottom) solutions for the Linearized Crooked Pipe.

	Runtime	Variance	FOM
HSM	0m 22s	$3.66 \cdot 10^{-4}$	124.2
UMC	41m 35s	$5.20 \cdot 10^{-5}$	7.7

Table 1: Median runtime, maximum variance, and figure of merit (FOM) for HSM and UMC calculations using 500 and 100 different pseudo-random number seeds, respectively.



(a)



(b)

Figure 14: The variance of the solution estimator for HSM (a) and UMC (b).

and the first to use variance reduction. We refer to our method as hybrid Second Moment (HSM).

HSM uses Monte Carlo (MC) instead of S_N to invert the transport operator. This approach avoids unphysical starburst patterns in the numerical solution, known as “ray effects”, which are caused by angular under-resolution in S_N methods. However, using MC introduces a variance issue, which we addressed by computing the deviation of the radiation intensity from isotropy. The magnitude of the source in the transformed problem is bounded, and the quality of the hybrid method solution is dramatically improved by a substantial reduction in the variance. Our analysis of the thick diffusion limit (TDL) provided a theoretical basis for the variance that we should expect with HSM, and we included a numerical demonstration of the hypothesized value. This motivated the inclusion of a transient heat conduction solve which, along with the second moment system solve, forms the deterministic component of HSM.

We showed that HSM performs well on a proxy problem from radiative transfer containing both optically-thick and optically-thin material. In this way, HSM is similar to thermal radiative transfer algorithms such as IMD or DDMC, which dramatically reduce the expense of implicit Monte Carlo (IMC) calculations in the TDL. However, IMD and DDMC make diffusion approximations, require phase-space partitioning, and have yet to be extended to reentrant elements. HSM does not make a diffusion approximation, does not partition phase-space, and does not require a non-reentrant mesh. Table 2 summarizes these differences.

	HSM	IMD or DDMC
Orders of magnitude speedups	✓	✓
No diffusion approximation	✓	×
No phase-space partitioning	✓	×
No issues with reentrant elements	✓	×
No iteration	×	✓
No linear solver	×	✓
No negative energy-weight	×	✓
Demonstrated in production calculations	×	✓

Table 2: Comparison of HSM and IMD or DDMC for selected qualities.

Future work on HSM could include investigating alternatives to the transient heat conduction solve, improvements to reduce the variance in the Linearized Crooked Pipe problem relative to unaccelerated Monte Carlo, comparisons in cost/accuracy to IMD and DDMC, extending HSM to thermal radiative transfer, and accommodating more physics such as line transport or Compton scattering in the MC solve and the deterministic low-order system. The anisotropy, nonlinearity, and frequency dependence of the Compton scattering kernel make it much more complicated than the isotropic, linear, frequency-independent scattering kernel considered in this manuscript. Nonetheless, the same two options used for the latter would also apply to the former: accounting for the contribution of the term either i) in the particle histories with scattering events, or ii) in the particle weights.

8. Acknowledgements

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